# CS201 LECTURE WISE MCQS 1 TO 22

# BY VU BSCS

# LECTURE 1

## Multiple-Choice Questions

1. **What is programming?**
   - A) A hobby
   - B) A precise sequence of steps to solve a problem
   - C) A type of software
   - D) An art form

   **Answer: B**
2. **Why is programming important according to Matthias Felleisen?**
   - A) It is only for high-tech programmers
   - B) It helps in developing analytical and problem-solving skills
   - C) It is the only way to create software
   - D) It requires advanced mathematics

   **Answer: B**
3. **Which skill is emphasized as crucial for programming?**
   - A) Memorization
   - B) Paying attention to detail
   - C) Graphic design
   - D) Networking

   **Answer: B**
4. **What does Alan Perlis sarcastically suggest about modern education?**
   - A) It encourages programming
   - B) It teaches critical thinking
   - C) It neglects planning and organization skills
   - D) It focuses too much on technology

   **Answer: C**
5. **When programming, one should remember that:**
   - A) Computers can interpret vague instructions
   - B) Computers are smart
   - C) Computers are stupid and do exactly what they are told
   - D) Programming is unnecessary in modern software

   **Answer: C**
6. **What is an important aspect of user interface design in programming?**
   - A) It should be complex
   - B) It should assume the user is tech-savvy
   - C) It should be user-friendly and self-explanatory
   - D) It should focus on aesthetics

   **Answer: C**

7. **What does the design recipe in programming compare to?**
    - A) Cooking
    - B) Playing soccer
    - C) Writing essays
    - D) Painting

    **Answer: B**
8. **What is a key factor in writing reusable code?**
    - A) Avoiding comments
    - B) Writing for a specific problem only
    - C) Thinking about future applications
    - D) Making it complex

    **Answer: C**
9. **Which of the following is NOT a skill needed for programming?**
    - A) Attention to detail
    - B) Analytical thinking
    - C) Ability to memorize code
    - D) Creative synthesis

    **Answer: C**
10. **In the programming design recipe, the first step is to:**
    - A) Write code directly
    - B) Analyze the problem statement
    - C) Create a user interface
    - D) Debug the program

    **Answer: B**
11. **Why is it important to comment code?**
    - A) It enhances performance
    - B) It makes the code look professional
    - C) It helps others and the programmer understand the code
    - D) It reduces memory usage

    **Answer: C**
12. **What is a common mistake programmers make regarding users?**
    - A) Underestimating their needs
    - B) Over-explaining features
    - C) Assuming users know everything
    - D) Ignoring user feedback

    **Answer: C**
13. **What should be included in the analysis of a problem statement?**
    - A) Only theoretical examples
    - B) Real-world applications only
    - C) Abstract essence and concrete examples
    - D) A detailed code

    **Answer: C**
14. **What is the significance of logical flow in programming?**
    - A) It makes the code lengthy
    - B) It ensures the program compiles successfully
    - C) It determines the correctness of the program's output

- D) It simplifies debugging

**Answer: C**

15. **How should programmers think about the computer's capabilities?**
    - A) Computers can infer meaning
    - B) Computers can think independently
    - C) Computers can only execute explicit instructions
    - D) Computers understand context

    **Answer: C**
16. **In programming, "creative synthesis" refers to:**
    - A) Combining existing ideas to form new solutions
    - B) Writing code without planning
    - C) Strictly following coding conventions
    - D) Avoiding user input

    **Answer: A**
17. **What does “evaluate and revise” mean in the design process?**
    - A) Ignore feedback
    - B) Test the program and improve it as needed
    - C) Write new code from scratch
    - D) Document the process only

    **Answer: B**
18. **The phrase "pay attention to detail" implies:**
    - A) Ignore minor errors
    - B) Focus only on major issues
    - C) Be thorough in analyzing and implementing code
    - D) Write code quickly

    **Answer: C**
19. **When developing a payroll system, which factor is essential?**
    - A) User feedback is irrelevant
    - B) Each category of employee should be considered
    - C) Only permanent employees should be included
    - D) Aesthetic design is paramount

    **Answer: B**
20. **What is a common analogy for a programmer's role?**
    - A) A musician
    - B) A cook
    - C) An architect
    - D) A scientist

    **Answer: C**
21. **Why should programmers avoid assuming user knowledge?**
    - A) Users always know more than programmers
    - B) It can lead to user frustration and poor interface design
    - C) Users are typically developers themselves
    - D) It's not important

    **Answer: B**
22. **What is one way to ensure code is understandable?**
    - A) Use complex terminology

- B) Avoid comments
- C) Use clear variable names and comments
- D) Write everything in one line

**Answer: C**

23. **The first part of programming involves:**
    - A) Compiling code
    - B) Debugging errors
    - C) Writing down the steps to solve a problem
    - D) Finalizing the user interface

    **Answer: C**
24. **What is the essence of programming, as suggested in the lecture?**
    - A) It's a tedious process
    - B) It's a creative form of problem-solving
    - C) It's solely for professionals
    - D) It's a mathematical exercise

    **Answer: B**
25. **The iterative nature of programming suggests that:**
    - A) One can write code without revisions
    - B) Programs should be developed in a linear fashion
    - C) Programs should be constantly evaluated and refined
    - D) Testing is unnecessary

    **Answer: C**

# LECTURE 2

## Multiple Choice Questions

1. **What is programming?**
    - A) Writing letters
    - B) A sequence of steps to solve a problem
    - C) Designing websites
    - D) None of the above

    **Answer:** B
2. **Which of the following is NOT a category of system software?**
    - A) Operating System
    - B) Device Drivers
    - C) Spreadsheets
    - D) Utilities

    **Answer:** C
3. **What is the primary function of an operating system?**
    - A) To create applications
    - B) To manage hardware resources
    - C) To perform calculations
    - D) To design user interfaces

    **Answer:** B

4. **What type of software helps in managing system resources?**
    - A) Application Software
    - B) System Software
    - C) Utility Software
    - D) Firmware

    **Answer:** C
5. **Which language was primarily used for developing the UNIX operating system?**
    - A) Java
    - B) Python
    - C) C
    - D) Fortran

    **Answer:** C
6. **What does a device driver do?**
    - A) Manages files
    - B) Communicates between the OS and hardware
    - C) Creates applications
    - D) None of the above

    **Answer:** B
7. **What is a compiler?**
    - A) A program that executes code line by line
    - B) A tool that translates the entire program into machine code
    - C) A type of software that manages devices
    - D) A program used for debugging

    **Answer:** B
8. **What is the function of a debugger?**
    - A) To compile the program
    - B) To manage files
    - C) To identify and correct logical errors
    - D) To load programs into memory

    **Answer:** C
9. **Which of the following is an example of application software?**
    - A) Operating System
    - B) Word Processor
    - C) Device Driver
    - D) Disk Utility

    **Answer:** B
10. **What are utilities used for?**
    - A) Writing code
    - B) Managing system resources
    - C) Creating user interfaces
    - D) None of the above

    **Answer:** B
11. **Which of the following is a characteristic of C language?**
    - A) It is a high-level language.
    - B) It is an interpreted language.
    - C) It does not support data types.

- D) It is primarily used for web development.

**Answer:** A

12. **What does the linker do?**
    - A) Compiles the code
    - B) Links object code with libraries
    - C) Loads programs into memory
    - D) Translates code line by line

    **Answer:** B
13. **Why is it important to comment code?**
    - A) To make it look pretty
    - B) To explain functionality to others
    - C) To increase execution speed
    - D) To reduce file size

    **Answer:** B
14. **What is the role of the loader?**
    - A) To translate code
    - B) To link code
    - C) To load programs into memory for execution
    - D) To edit code

    **Answer:** C
15. **Which of the following is a disadvantage of using an interpreter?**
    - A) Executes code too quickly
    - B) Cannot provide error feedback
    - C) Executes line by line
    - D) Needs more memory

    **Answer:** C
16. **In which year was the C standard approved by ANSI?**
    - A) 1980
    - B) 1983
    - C) 1989
    - D) 1995

    **Answer:** C
17. **What is the essence of programming as described by Steve Summit?**
    - A) It is about using advanced tools.
    - B) It is merely telling the computer what to do.
    - C) It is primarily for scientists and engineers.
    - D) It requires no planning.

    **Answer:** B
18. **Which software manages hardware allocation?**
    - A) Application Software
    - B) Device Drivers
    - C) Operating System
    - D) Utility Software

    **Answer:** C
19. **What is one important skill needed for programming?**
    - A) Writing long documents

- B) Paying attention to detail
- C) Designing graphics
- D) None of the above

**Answer:** B

20. **What does the term "fragmentation" refer to in disk management?**
    - A) Combining files
    - B) Scattering of data across the disk
    - C) Deleting files
    - D) Compressing files

    **Answer:** B
21. **What is the function of a text editor in programming?**
    - A) To enhance visual appearance
    - B) To write and save plain text code
    - C) To execute programs
    - D) To debug code

    **Answer:** B
22. **What does "C" language stand for in programming?**
    - A) Code
    - B) Comprehensive
    - C) Compiler
    - D) None of the above

    **Answer:** D
23. **Who developed the C programming language?**
    - A) Ken Thompson
    - B) Dennis Ritchie
    - C) Martin Richards
    - D) Alan Turing

    **Answer:** B
24. **What is an example of a utility program?**
    - A) Microsoft Word
    - B) Disk Defragmenter
    - C) Web Browser
    - D) Video Player

    **Answer:** B
25. **Which of the following statements is TRUE about C?**
    - A) It is a type-less language.
    - B) It is exclusively for web programming.
    - C) It uses data types and is structured.
    - D) It is no longer used.

    **Answer:** C

## LECTURE 3

**MCQs**

1. **What is the output of the following code?**

```
c
Copy code
cout << "Hello, World!";
```

a) Hello, World!
b) "Hello, World!"
c) Error
d) Hello World
**Answer:** a) Hello, World!

2. **Which of the following is a valid variable name in C?** a) 1variable
b) variable_1
c) variable-1
c) variable 1
**Answer:** b) variable_1
3. **What does the following declaration do?** `int x;` a) Defines x as a float
b) Reserves 4 bytes for an integer variable
c) Initializes x to 0
d) None of the above
**Answer:** b) Reserves 4 bytes for an integer variable
4. **Which operator is used for assignment in C?** a) =
b) ==
c) :=
d) =>
**Answer:** a) =
5. **What will be the output of** `cout << 10 + 20 * 2;`**?** a) 60
b) 50
c) 40
d) 70
**Answer:** b) 50
6. **Which of the following is NOT a data type in C?** a) int
b) float
c) string
d) char
**Answer:** c) string
7. **What does the modulus operator (%) return?** a) Division result
b) Remainder of division
c) Sum of two numbers
d) Absolute value
**Answer:** b) Remainder of division
8. **What is the size of an** `int` **data type in C on most systems?** a) 2 bytes
b) 4 bytes
c) 8 bytes
d) 1 byte
**Answer:** b) 4 bytes

9. **Which of the following correctly declares a float variable?** a) float x;
   b) int x;
   c) double x;
   d) char x;
   **Answer:** a) float x;
10. **What is the purpose of the `main()` function in a C program?** a) To declare variables
    b) To execute the program
    c) To define functions
    d) None of the above
    **Answer:** b) To execute the program
11. **Which of the following will cause a syntax error?** a) cout << "Hello";
    b) int x = 10;
    c) float y 5.5;
    d) double z;
    **Answer:** c) float y 5.5;
12. **What is the output of `cout << "x = " << x;` if `x = 10;`?** a) x = 10
    b) "x = 10"
    c) Error
    d) x = "10"
    **Answer:** a) x = 10
13. **Which operator has the highest precedence in C?** a) +
    b) -
    c) *
    d) /
    **Answer:** c) *
14. **How do you declare a character variable?** a) char x;
    b) int x;
    c) string x;
    d) character x;
    **Answer:** a) char x;
15. **Which of the following is a floating-point data type?** a) int
    b) char
    c) float
    d) short
    **Answer:** c) float
16. **What will be the output of the following code?**

```
c
Copy code
int x = 5, y = 2;
cout << x % y;
```

    a) 2
    b) 1
    c) 3
    d) 0
    **Answer:** b) 1

17. **What is the correct way to include a header file in C?** a) #include <header.h>
    b) include <header.h>
    c) #include "header.h"
    d) a and c
    **Answer:** d) a and c
18. **Which statement is true about C variables?** a) They can start with a number.
    b) They are case-sensitive.
    c) They can contain spaces.
    d) They must be declared after usage.
    **Answer:** b) They are case-sensitive.
19. **What is the output of `cout << 15 / 2;`?** a) 7.5
    b) 7
    c) 8
    d) 15
    **Answer:** b) 7
20. **What will be stored in `char x = 'a';`?** a) The character a
    b) The integer 97
    c) Both a and b
    d) Error
    **Answer:** c) Both a and b
21. **What does `#define PI 3.14` do?** a) Declares a variable PI
    b) Creates a constant named PI
    c) Creates a function named PI
    d) None of the above
    **Answer:** b) Creates a constant named PI
22. **Which of the following statements correctly assigns a value to a variable?** a) x := 10;
    b) x = 10;
    c) 10 = x;
    d) x => 10;
    **Answer:** b) x = 10;
23. **How do you declare an array of integers in C?** a) int arr[];
    b) int arr[10];
    c) int arr = new int[10];
    d) Both a and b
    **Answer:** b) int arr[10];
24. **Which operator is used for incrementing a variable by one?** a) +1
    b) ++
    c) +=
    d) --
    **Answer:** b) ++
25. **What will the following code output?**

```
c
Copy code
int x = 5;
x++;
cout << x;
```

a) 5
b) 6
c) 7
d) 4
**Answer:** b) 6

26. **Which of the following can store a decimal value?** a) int
    b) char
    c) float
    d) short
    **Answer:** c) float
27. **Which is a valid comment in C?** a) // This is a comment
    b) /* This is a comment */
    c) Both a and b
    d) None of the above
    **Answer:** c) Both a and b
28. **What is the output of `cout << "10" + 5;`?** a) 15
    b) Error
    c) "105"
    d) 105
    **Answer:** b) Error
29. **Which statement correctly prints a variable?** a) cout.print(x);
    b) print(cout, x);
    c) cout << x;
    d) System.out(x);
    **Answer:** c) cout << x;
30. **What is the purpose of the `return 0;` statement in `main()`?** a) To terminate the program
    b) To indicate successful execution
    c) To print 0
    d) None of the above
    **Answer:** b) To indicate successful execution

# LECTURE 4

## MCQs

1. **What is the correct data type for storing ages in whole numbers?**
    - A) float
    - B) int
    - C) double
    - D) char

    **Answer: B) int**
2. **What operator is used for assignment in C++?**
    - A) =

- B) ==
- C) :=
- D) =>

**Answer: A) =**

3. **In the program, how many student ages are collected?**
   - A) 5
   - B) 8
   - C) 10
   - D) 15

   **Answer: C) 10**
4. **What will be the output of `cout << "The average age of the students is : " << AverageAge;`?**
   - A) The average age of the students is : 0
   - B) The average age of the students is : (value of AverageAge)
   - C) The average age of the students is : undefined
   - D) The average age of the students is : 10

   **Answer: B) The average age of the students is : (value of AverageAge)**
5. **Which operator is used to calculate the modulus?**
   - A) /
   - B) *
   - C) %
   - D) +

   **Answer: C) %**
6. **What does the expression `TotalAge = age1 + age2 + ... + age10;` do?**
   - A) Calculates the average age
   - B) Calculates the total age
   - C) Assigns the age of the first student
   - D) Prompts user input

   **Answer: B) Calculates the total age**
7. **What type of division is performed in `AverageAge = TotalAge / 10;` if both TotalAge and 10 are integers?**
   - A) Floating-point division
   - B) Integer division
   - C) Complex division
   - D) None of the above

   **Answer: B) Integer division**
8. **Which line prompts the user to enter the age of the first student?**
   - A) `cout << "Please enter the age of student 1: ";`
   - B) `cin >> age1;`
   - C) `cout << "Enter age: ";`
   - D) `cout << "First student's age: ";`

   **Answer: A) `cout << "Please enter the age of student 1: ";`**
9. **How many variables are declared to store the ages of students?**
   - A) 5
   - B) 10
   - C) 15

- D) 20

**Answer: B) 10**

10. **What type of output is displayed when the program runs successfully?**
    - A) A string
    - B) An integer
    - C) A float
    - D) A character

    **Answer: A) A string**
11. **Which statement will correctly take an integer input from the user?**
    - A) `cin << age1;`
    - B) `cin = age1;`
    - C) `cin >> age1;`
    - D) `cin < age1;`

    **Answer: C) `cin >> age1;`**
12. **What does the operator `*` represent in C++?**
    - A) Addition
    - B) Subtraction
    - C) Multiplication
    - D) Division

    **Answer: C) Multiplication**
13. **If the total age is 123, what will be the average age when divided by 10?**
    - A) 12
    - B) 12.3
    - C) 13
    - D) 12.0

    **Answer: A) 12**
14. **What will happen if the program attempts to divide by zero?**
    - A) It will compile successfully
    - B) It will run but return 0
    - C) It will cause a runtime error
    - D) It will produce an infinite loop

    **Answer: C) It will cause a runtime error**
15. **Which of the following correctly represents the area of a circle formula in C++?**
    - A) `area = pi * radius;`
    - B) `area = 2 * pi * radius;`
    - C) `area = pi * radius * radius;`
    - D) `area = radius / 2;`

    **Answer: C) `area = pi * radius * radius;`**
16. **What is the result of `age1 % 10` if age1 is 25?**
    - A) 2
    - B) 5
    - C) 0
    - D) 25

    **Answer: B) 5**
17. **What kind of values does the `float` data type store?**
    - A) Only whole numbers

- B) Only character values
- C) Decimal numbers
- D) Boolean values

**Answer: C) Decimal numbers**

18. **What will the program display if the total age of students is 173?**
    - A) 17.3
    - B) 17
    - C) 18
    - D) 173

    **Answer: B) 17**
19. **Which operator has the highest precedence in the expression `a + b * c`?**
    - A) +
    - B) *
    - C) Both have equal precedence
    - D) None of the above

    **Answer: B) ***
20. **In the expression `x = (a * x) + (b * y) + c * (z * z)`, what does the parentheses do?**
    - A) They change the data type
    - B) They control the order of evaluation
    - C) They concatenate strings
    - D) They declare variables

    **Answer: B) They control the order of evaluation**
21. **What will the output be if `TotalAge` is 50 and `AverageAge` is calculated as `TotalAge / 10`?**
    - A) 5
    - B) 50
    - C) 0
    - D) 10

    **Answer: A) 5**
22. **What is the purpose of using `cout` in a C++ program?**
    - A) To take user input
    - B) To display output
    - C) To store values
    - D) To perform calculations

    **Answer: B) To display output**
23. **Which of the following is a correct way to comment in C++?**
    - A) // This is a comment
    - B) /* This is a comment */
    - C) # This is a comment
    - D) Both A and B

    **Answer: D) Both A and B**
24. **What does `cin >> age2;` do?**
    - A) Assigns the value of age2 to the input stream
    - B) Takes input from the user and assigns it to age2
    - C) Displays the value of age2

- D) Declares age2 as a variable

**Answer: B) Takes input from the user and assigns it to age2**

25. **Which library is included at the beginning of the program for input/output operations?**
    - A) <cmath>
    - B) <iostream.h>
    - C) <stdio.h>
    - D) <cstdlib>

    **Answer: B) <iostream.h>**
26. **What will `cout << "The circumference of the circle is : " << circumference;` display if circumference is 31.4?**
    - A) The circumference of the circle is : 0
    - B) The circumference of the circle is : 31.4
    - C) The circumference of the circle is : (value of circumference)
    - D) The circumference of the circle is : undefined

    **Answer: B) The circumference of the circle is : 31.4**
27. **What is the first step in solving a programming problem according to the tips?**
    - A) Write the code
    - B) Test the code
    - C) Analyze the problem
    - D) Compile the code

    **Answer: C) Analyze the problem**
28. **What will happen if `TotalAge` is declared but not initialized?**
    - A) It will cause a compile error
    - B) It will have a default value of 0
    - C) It will contain garbage values
    - D) It will produce a warning

    **Answer: C) It will contain garbage values**
29. **Which of the following best describes the use of parentheses in C++ expressions?**
    - A) To make code look clean
    - B) To declare variables
    - C) To control the order of operations
    - D) To perform addition

    **Answer: C) To control the order of operations**
30. **What is the expected output if the user enters the ages 10, 20, 30, 40, 50, 60, 70, 80, 90, 100?**
    - A) The average age of the students is : 50
    - B) The average age of the students is : 55
    - C) The average age of the students is : 60
    - D) The average age of the students is : 70

    **Answer: B) The average age of the students is : 55**

## LECTURE 5

### MCQs on Conditional Statements in C++

1. **What is the correct syntax for an if statement in C++?**
   - A) `if condition { statements }`
   - B) `if (condition) statements`
   - C) `if (condition) { statements }`
   - D) `if { condition } statements`
   - **Answer: C**
2. **Which of the following is a relational operator in C++?**
   - A) `=`
   - B) `==`
   - C) `=>`
   - D) `!`
   - **Answer: B**
3. **What does the `!=` operator do?**
   - A) Checks if two values are equal
   - B) Checks if two values are not equal
   - C) Assigns a value
   - D) Compares two variables
   - **Answer: B**
4. **Which statement is true about the else block?**
   - A) It always executes.
   - B) It executes if the if condition is true.
   - C) It executes if the if condition is false.
   - D) It cannot contain statements.
   - **Answer: C**
5. **What will be the output of the following code?**

```
cpp
Copy code
int x = 5;
if (x > 3) {
    cout << "Hello";
} else {
    cout << "World";
}
```

   - A) Hello
   - B) World
   - C) Hello World
   - D) No output
   - **Answer: A**
6. **Which of the following represents a logical AND operation?**
   - A) `||`
   - B) `&&`
   - C) `!`
   - D) `==`
   - **Answer: B**
7. **What is the purpose of the `!` operator in C++?**

- A) To assign a value
- B) To check equality
- C) To negate a condition
- D) To compare two values
- **Answer: C**

8. **What will be the output of the following code?**

```
cpp
Copy code
int a = 10, b = 20;
if (a < b && a > 5) {
    cout << "True";
} else {
    cout << "False";
}
```

- A) True
- B) False
- C) 10
- D) 20
- **Answer: A**

9. **Which of the following is NOT a relational operator?**
   - A) `>`
   - B) `<`
   - C) `=`
   - D) `!=`
   - **Answer: C**
10. **How do you write an if statement that checks if `x` is equal to 10?**
    - A) `if (x == 10)`
    - B) `if (x = 10)`
    - C) `if x = 10`
    - D) `if (x = 10)`
    - **Answer: A**
11. **What will the following code output?**

```
cpp
Copy code
int age = 18;
if (age >= 18) {
    cout << "Adult";
} else {
    cout << "Minor";
}
```

- A) Minor
- B) Adult
- C) Error
- D) No output
- **Answer: B**

12. **Which keyword is used to create a block of code that executes when the condition is false?**
    - A) else
    - B) elseif
    - C) switch
    - D) default
    - **Answer: A**
13. **What is the output of the following code?**

```
cpp
Copy code
int x = 3;
if (x > 5) {
    cout << "Greater";
} else {
    cout << "Smaller";
}
```

    - A) Greater
    - B) Smaller
    - C) No output
    - D) Error
    - **Answer: B**
14. **What does the >= operator check for?**
    - A) Greater than
    - B) Less than
    - C) Greater than or equal to
    - D) Equal to
    - **Answer: C**
15. **Which statement about if-else if ladder is TRUE?**
    - A) It can have multiple else statements.
    - B) It is used to check multiple conditions.
    - C) It can only check two conditions.
    - D) It cannot be nested.
    - **Answer: B**
16. **What will be the output if x is 10 and y is 10 in the following code?**

```
cpp
Copy code
if (x > y) {
    cout << "X";
} else if (x < y) {
    cout << "Y";
} else {
    cout << "Equal";
}
```

    - A) X
    - B) Y

- C) Equal
- D) No output
- **Answer: C**

17. **Which of the following is a valid if statement?**
    - A) `if x > 5`
    - B) `if (x > 5) {}`
    - C) `if (x > 5) then {}`
    - D) `if (x > 5) {;}`
    - **Answer: B**
18. **What will happen if you write `if (x = 10)` instead of `if (x == 10)`?**
    - A) Syntax error
    - B) x will be assigned 10
    - C) It will compare x and 10
    - D) Logical error
    - **Answer: B**
19. **How can you write an if statement that checks if x is less than 10 or greater than 20?**
    - A) `if (x < 10 || x > 20)`
    - B) `if (x < 10 and x > 20)`
    - C) `if (x < 10 && x > 20)`
    - D) `if (x < 10 or x > 20)`
    - **Answer: A**
20. **What is the output of the following code?**

```
cpp
Copy code
int n = 0;
if (n) {
    cout << "True";
} else {
    cout << "False";
}
```

    - A) True
    - B) False
    - C) Error
    - D) No output
    - **Answer: B**
21. **Which operator is used to compare if two values are equal?**
    - A) `=`
    - B) `==`
    - C) `!=`
    - D) `===`
    - **Answer: B**
22. **What is the result of the expression `5 < 3 || 4 > 2`?**
    - A) True
    - B) False
    - C) Error

- D) Undefined
- **Answer: A**

23. **How can you negate a condition using logical NOT?**
    - A) `if (!condition)`
    - B) `if (condition!)`
    - C) `if (!condition!)`
    - D) `if (not condition)`
    - **Answer: A**
24. **Which statement is used to provide alternative code paths in C++?**
    - A) switch
    - B) loop
    - C) if-else
    - D) function
    - **Answer: C**
25. **What will the following code print if `x = 4`?**

```
cpp
Copy code
if (x == 5) {
    cout << "Five";
} else {
    cout << "Not Five";
}
```

    - A) Five
    - B) Not Five
    - C) Error
    - D) No output
    - **Answer: B**
26. **In C++, which of the following is NOT a valid condition?**
    - A) `x > y`
    - B) `x = y`
    - C) `x == y`
    - D) `x != y`
    - **Answer: B**
27. **Which is the correct way to check multiple conditions?**
    - A) `if (a && b && c)`
    - B) `if (a or b or c)`
    - C) `if (a & b & c)`
    - D) `if (a || b || c)`
    - **Answer: D**
28. **What will happen if you use an `else` without a preceding `if`?**
    - A) It will compile successfully.
    - B) It will cause a syntax error.
    - C) It will execute the else block.
    - D) It will result in a logical error.
    - **Answer: B**

29. **What is the primary use of the if statement?**
    - A) To repeat a block of code
    - B) To declare variables
    - C) To make decisions based on conditions
    - D) To exit a program
    - **Answer: C**
30. **Which statement is true about the condition in an if statement?**
    - A) It must be an integer.
    - B) It must be a boolean expression.
    - C) It can be a string.
    - D) It can be a float.
    - **Answer: B**
31. **What will be the output of the following code?**

```
cpp
Copy code
int x = 10;
if (x < 5) {
    cout << "Small";
} else if (x > 15) {
    cout << "Large";
} else {
    cout << "Medium";
}
```

    - A) Small
    - B) Large
    - C) Medium
    - D) No output
    - **Answer: C**
32. **What does the expression `x < 5 && x > 10` evaluate to?**
    - A) Always true
    - B) Always false
    - C) True if x is between 5 and 10
    - D) True if x is less than 5 or greater than 10
    - **Answer: B**
33. **Which of the following correctly represents an if statement with a nested if?**
    - A)

```
cpp
Copy code
if (condition1)
    if (condition2)
        // statements
```

    - B)

```
cpp
Copy code
if (condition1) {
```

```
    if (condition2)
        // statements
}
```

- C) Both A and B
- D) None of the above
- **Answer: C**

34. **What will the output be for the following code?**

```
cpp
Copy code
int a = 10;
if (a > 0) {
    cout << "Positive";
} else {
    cout << "Negative";
}
```

- A) Positive
- B) Negative
- C) Zero
- D) Error
- **Answer: A**

35. **Which operator can be used to combine multiple conditions in an if statement?**
    - A) `==`
    - B) `&&`
    - C) `!=`
    - D) `+`
    - **Answer: B**
36. **What is the output of this code if `x` is 7?**

```
cpp
Copy code
if (x < 5) {
    cout << "Less than 5";
} else if (x == 7) {
    cout << "Seven";
} else {
    cout << "Greater than 5";
}
```

- A) Less than 5
- B) Seven
- C) Greater than 5
- D) No output
- **Answer: B**

37. **Which of the following statements is used to execute a block of code when a specific condition is met?**
    - A) loop
    - B) if

- C) case
- D) function
- **Answer: B**

38. **What does the code below output if `num = 0`?**

```
cpp
Copy code
if (num) {
    cout << "Non-zero";
} else {
    cout << "Zero";
}
```

- A) Non-zero
- B) Zero
- C) Error
- D) No output
- **Answer: B**

39. **How do you create a condition that checks if `x` is NOT equal to `5`?**
    - A) `if (x != 5)`
    - B) `if (x =! 5)`
    - C) `if (x not 5)`
    - D) `if (x == 5)`
    - **Answer: A**
40. **What is the output of the following code?**

```
cpp
Copy code
int x = 20, y = 15;
if (x > y) {
    cout << "X is greater";
} else {
    cout << "Y is greater";
}
```

- A) X is greater
- B) Y is greater
- C) Error
- D) No output
- **Answer: A**

# LECTURE 6

## MCQs on C++ Loops

1. **What is the primary purpose of a loop in programming?**
    - A) To store data
    - B) To execute a block of code multiple times

- C) To make decisions
- D) To define functions
- **Answer:** B

2. **Which of the following is a type of loop in C++?**
   - A) if-else
   - B) switch
   - C) while
   - D) function
   - **Answer:** C
3. **What will be the output of the following code?**

```
cpp
Copy code
int sum = 0, i = 1;
while (i <= 5) {
    sum += i;
    i++;
}
cout << sum;
```

   - A) 10
   - B) 15
   - C) 5
   - D) 20
   - **Answer:** B
4. **What is the output of the following code?**

```
cpp
Copy code
int count = 0;
while (count < 3) {
    cout << count << " ";
    count++;
}
```

   - A) 0 1 2
   - B) 1 2 3
   - C) 0 1 2 3
   - D) Infinite loop
   - **Answer:** A
5. **What happens if the condition of a while loop is always true?**
   - A) The loop will execute once.
   - B) The loop will never execute.
   - C) An infinite loop occurs.
   - D) The program will crash.
   - **Answer:** C
6. **Which loop is guaranteed to execute at least once?**
   - A) for loop
   - B) while loop

- C) do-while loop
- D) infinite loop
- **Answer:** C

7. **What keyword is used to exit a loop prematurely?**
   - A) continue
   - B) exit
   - C) break
   - D) return
   - **Answer:** C
8. **In C++, which of the following is true about the do-while loop?**
   - A) It checks the condition before executing the loop body.
   - B) It can execute zero times.
   - C) It always executes at least once.
   - D) It cannot be nested.
   - **Answer:** C
9. **What will be the output of the following code?**

```
cpp
Copy code
for (int i = 0; i < 5; i++) {
    if (i == 3) {
        break;
    }
    cout << i << " ";
}
```

   - A) 0 1 2
   - B) 0 1 2 3
   - C) 0 1 2 3 4
   - D) Infinite loop
   - **Answer:** A
10. **Which of the following correctly initializes a while loop?**
    - A) while (i = 0)
    - B) while (i == 0)
    - C) while (i < 0)
    - D) while (i != 0)
    - **Answer:** B
11. **Which of the following statements is true about the for loop?**
    - A) It must have a condition.
    - B) It cannot be used for infinite loops.
    - C) It is always executed at least once.
    - D) It is typically used for iterating over a range.
    - **Answer:** D
12. **What is the output of the following code?**

```
cpp
Copy code
int i = 0;
while (i < 5) {
```

```
    cout << i++;
}
```

- A) 01234
- B) 12345
- C) 0 1 2 3 4
- D) 1 2 3 4 5
- **Answer:** A

13. **What will happen if you forget to update the variable in a while loop?**
    - A) The program will compile successfully.
    - B) The loop will terminate immediately.
    - C) An infinite loop may occur.
    - D) The program will throw an error.
    - **Answer:** C
14. **Which statement is used to skip the current iteration of a loop?**
    - A) stop
    - B) continue
    - C) break
    - D) skip
    - **Answer:** B
15. **How do you correctly write a for loop in C++?**
    - A) for (int i = 0; i < 10; i++) {}
    - B) for (int i = 0; i < 10) {}
    - C) for (i < 10; i++) {}
    - D) for (int i = 0; i++) {}
    - **Answer:** A
16. **Which of the following will result in an infinite loop?**
    - A) for (int i = 0; i < 10; i++)
    - B) while (true)
    - C) do {}
    - D) Both B and C
    - **Answer:** D
17. **What is the output of the following code?**

```
cpp
Copy code
int sum = 0;
for (int i = 1; i <= 10; i += 2) {
    sum += i;
}
cout << sum;
```

- A) 30
- B) 25
- C) 20
- D) 15
- **Answer:** B

18. **What will be the output of the following code?**

```
cpp
Copy code
int i = 1;
do {
    cout << i << " ";
    i++;
} while (i < 1);
```

- A) No output
- B) 1
- C) 1 2
- D) Infinite loop
- **Answer:** B

19. **Which loop can be used to iterate over an array?**
    - A) if-else
    - B) for loop
    - C) do-while
    - D) Both B and C
    - **Answer:** D
20. **Which of the following correctly demonstrates a nested loop?**
    - A) for (int i = 0; i < 10; for (int j = 0; j < 5; j++))
    - B) for (int i = 0; i < 10; i++) { for (int j = 0; j < 5; j++) {} }
    - C) for (int i = 0; i < 10) { for (int j = 0; j < 5; j++) {} }
    - D) Both A and C
    - **Answer:** B
21. **What will be the output of the following code?**

```
cpp
Copy code
for (int i = 0; i < 3; i++) {
    for (int j = 0; j < 2; j++) {
        cout << i << j << " ";
    }
}
```

    - A) 00 01 10 11 20 21
    - B) 00 01 10 20
    - C) 00 01 02 11 12
    - D) 00 01 10 11 20 21
    - **Answer:** D
22. **What does the `continue` statement do inside a loop?**
    - A) Exits the loop immediately.
    - B) Skips the rest of the loop body for the current iteration.
    - C) Causes an error.
    - D) Restarts the loop from the beginning.
    - **Answer:** B
23. **Which of the following conditions will cause a while loop to terminate?**
    - A) The condition is true.
    - B) The condition is false.

- C) Both true and false.
- D) There is no termination condition.
- **Answer:** B

24. **What will be the output of the following code?**

```
cpp
Copy code
int x = 10;
do {
    cout << x << " ";
    x--;
} while (x > 0);
```

- A) 10 9 8 7 6 5 4 3 2 1
- B) 10 9 8 7 6 5 4 3 2 1 0
- C) 10 9 8 7 6 5 4 3 2 1 0 -1
- D) No output
- **Answer:** A

25. **How can you initialize a variable inside a for loop?**
    - A) Only outside the loop
    - B) Only inside the loop
    - C) Both inside and outside the loop
    - D) It cannot be initialized
    - **Answer:** C
26. **What will be the result of the following code?**

```
cpp
Copy code
int i = 1;
while (i < 5) {
    i += 2;
    cout << i << " ";
}
```

- A) 1 2 3 4
- B) 2 4 6
- C) 3 5
- D) 3 5 7
- **Answer:** C

27. **Which statement can be used to end a loop and transfer control to the next statement outside the loop?**
    - A) return
    - B) exit
    - C) break
    - D) end
    - **Answer:** C
28. **What will be the output of the following code?**

```
cpp
```

```
Copy code
for (int i = 0; i < 5; i++) {
    if (i == 2) {
        continue;
    }
    cout << i << " ";
}
```

- A) 0 1 2 3 4
- B) 0 1 3 4
- C) 1 2 3 4
- D) 2 3 4
- **Answer:** B

29. **What is the syntax of the while loop?**
    - A) while condition { statements; }
    - B) while (condition) statements;
    - C) while { condition; statements; }
    - D) while(condition): statements;
    - **Answer:** A
30. **In a for loop, where do you typically place the increment expression?**
    - A) At the start of the loop
    - B) Inside the loop body
    - C) In the loop header
    - D) It can be placed anywhere
    - **Answer:** C
31. **What does the following code snippet print?**

```
cpp
Copy code
for (int i = 1; i <= 3; i++) {
    for (int j = 1; j <= 2; j++) {
        cout << i * j << " ";
    }
}
```

- A) 1 2 2 3 3 6
- B) 1 2 1 2 1 2
- C) 1 2 2 4 3 6
- D) 1 1 1 1
- **Answer:** C

32. **What type of loop would be most appropriate to process all elements of an array?**
    - A) while loop
    - B) do-while loop
    - C) for loop
    - D) if-else statement
    - **Answer:** C
33. **How can you avoid infinite loops?**
    - A) Always use a break statement
    - B) Ensure the loop condition will eventually become false

- C) Use a return statement
- D) None of the above
- **Answer:** B

34. **What is the output of this code?**

```
cpp
Copy code
int i = 0;
while (i < 4) {
    cout << "Hello ";
    i++;
}
```

- A) Hello Hello Hello
- B) Hello Hello Hello Hello
- C) Hello
- D) Infinite loop
- **Answer:** B

35. **Which of the following is an incorrect way to write a loop?**
    - A) for (int i = 0; i < 10; i++)
    - B) while (x > 0)
    - C) do { } while (true);
    - D) for (int i = 0; i < 10) { }
    - **Answer:** D

# LECTURE 7

## MCQs on C++ Loops

1. **What is the primary purpose of a loop in programming?**
    - A) To execute a block of code once
    - B) To execute a block of code repeatedly
    - C) To terminate the program
    - D) To define variables
    - **Correct Answer:** B
2. **In a while loop, when is the condition checked?**
    - A) Before the loop starts
    - B) After the loop body executes
    - C) During the loop execution
    - D) At the end of the program
    - **Correct Answer:** A
3. **What is a do-while loop guaranteed to do?**
    - A) Execute at least once
    - B) Never execute
    - C) Execute indefinitely
    - D) Execute only when the condition is true

- **Correct Answer:** A

4. **Which statement about the for loop is true?**
   - A) It can only increment variables
   - B) It is not suitable for a known number of repetitions
   - C) It consists of initialization, condition, and increment/decrement
   - D) It must always execute at least once
   - **Correct Answer:** C
5. **What will the following for loop print?**

```
cpp
Copy code
for (int i = 0; i < 5; i++) {
    cout << i << " ";
}
```

   - A) 0 1 2 3 4
   - B) 1 2 3 4 5
   - C) 0 1 2 3 4 5
   - D) 1 2 3 4
   - **Correct Answer:** A
6. **Which operator is used to increment a variable by 1?**
   - A) --
   - B) +=
   - C) ++
   - D) -=
   - **Correct Answer:** C
7. **What is the output of the following code?**

```
cpp
Copy code
int x = 5;
cout << ++x;
```

   - A) 5
   - B) 6
   - C) 7
   - D) 0
   - **Correct Answer:** B
8. **Which loop will execute an infinite number of times if the condition is always true?**
   - A) for loop
   - B) while loop
   - C) do-while loop
   - D) All of the above
   - **Correct Answer:** D
9. **In a do-while loop, when is the condition evaluated?**
   - A) Before the loop
   - B) After the loop body
   - C) During the loop execution

- D) None of the above
- **Correct Answer:** B

10. **What does the following code snippet do?**

```
cpp
Copy code
for (int i = 10; i >= 0; i--) {
    cout << i << " ";
}
```

- A) Counts up from 10 to 0
- B) Counts down from 10 to 0
- C) Prints only 10
- D) Infinite loop
- **Correct Answer:** B

11. **Which of the following is NOT a valid loop in C++?**
    - A) for
    - B) while
    - C) until
    - D) do-while
    - **Correct Answer:** C
12. **How many times will the following loop execute?**

```
cpp
Copy code
int i = 0;
while (i < 5) {
    i++;
}
```

- A) 4
- B) 5
- C) 6
- D) Infinite
- **Correct Answer:** B

13. **What will be the output of this code?**

```
cpp
Copy code
for (int i = 0; i < 3; i++) {
    for (int j = 0; j < 2; j++) {
        cout << "*";
    }
}
```

- A) ***
- B) ******
- C) ***
- D) *****
- **Correct Answer:** B

14. **What is the difference between pre-increment and post-increment?**
    - A) Pre-increment increases the value before use; post-increment increases after use.
    - B) There is no difference.
    - C) Post-increment is faster than pre-increment.
    - D) Pre-increment can only be used in loops.
    - **Correct Answer:** A
15. **What is the purpose of the compound assignment operator +=?**
    - A) To assign a new value
    - B) To add a value and assign it to a variable
    - C) To subtract a value and assign it
    - D) To increment a variable by 2
    - **Correct Answer:** B
16. **Which of the following will not terminate the loop?**
    - A) Changing the loop variable incorrectly
    - B) Setting the condition to false
    - C) Properly incrementing the loop variable
    - D) Using the break statement
    - **Correct Answer:** A
17. **In C++, what keyword is used to exit a loop immediately?**
    - A) continue
    - B) stop
    - C) exit
    - D) break
    - **Correct Answer:** D
18. **What will be the output of the following code?**

```
cpp
Copy code
int sum = 0;
for (int i = 1; i <= 5; i++) {
    sum += i;
}
cout << sum;
```

    - A) 10
    - B) 15
    - C) 20
    - D) 5
    - **Correct Answer:** B
19. **What does the following code do?**

```
cpp
Copy code
for (int i = 0; i < 5; i++) {
    cout << i << " ";
}
```

    - A) Prints numbers from 1 to 5

- B) Prints numbers from 0 to 4
- C) Prints numbers from 0 to 5
- D) Prints numbers from 1 to 4
- **Correct Answer:** B

20. **In the context of loops, what does an "infinite loop" mean?**
    - A) A loop that executes a set number of times
    - B) A loop that never ends
    - C) A loop that is always true
    - D) Both B and C
    - **Correct Answer:** D
21. **Which of the following is used to control the flow of a loop?**
    - A) Break
    - B) Continue
    - C) Both A and B
    - D) None of the above
    - **Correct Answer:** C
22. **What is the role of the `continue` statement in loops?**
    - A) To exit the loop
    - B) To skip the current iteration and proceed to the next
    - C) To pause the loop
    - D) To restart the loop
    - **Correct Answer:** B
23. **How can you ensure that a do-while loop executes at least once?**
    - A) Place the condition first
    - B) Use a while loop instead
    - C) Use a do-while structure
    - D) Change the loop variable before the condition
    - **Correct Answer:** C
24. **What will be the output of the following code?**

```
cpp
Copy code
for (int i = 0; i < 3; i++) {
    for (int j = 0; j < 3; j++) {
        cout << "X";
    }
    cout << endl;
}
```

    - A) A single line of X's
    - B) A square of X's
    - C) Three lines of X's
    - D) Three columns of X's
    - **Correct Answer:** B
25. **What does the expression `i += 2` do?**
    - A) Increments i by 1
    - B) Increments i by 2
    - C) Decrements i by 2

- D) Resets i to 2
- **Correct Answer:** B

26. **Which of the following correctly describes the structure of a for loop?**
    - A) for (initialization; condition; increment)
    - B) for (condition; increment; initialization)
    - C) for (increment; initialization; condition)
    - D) for (initialization; increment; condition)
    - **Correct Answer:** A
27. **What is the output of this code snippet?**

```
cpp
Copy code
int x = 0;
for (int i = 0; i < 3; i++) {
    x++;
}
cout << x;
```

    - A) 2
    - B) 3
    - C) 0
    - D) Infinite
    - **Correct Answer:** B
28. **Which loop is typically used when the number of iterations is unknown?**
    - A) for loop
    - B) while loop
    - C) do-while loop
    - D) Both B and C
    - **Correct Answer:** D
29. **What will happen if the loop condition never becomes false?**
    - A) The loop will exit normally
    - B) The program will crash
    - C) The loop will run infinitely
    - D) The loop will print an error message
    - **Correct Answer:** C
30. **What is the effect of the following code?**

```
cpp
Copy code
int i = 10;
while (i > 0) {
    cout << i << " ";
    i--;
}
```

    - A) Prints numbers from 1 to 10
    - B) Prints numbers from 10 to 1
    - C) Infinite loop
    - D) No output

- **Correct Answer:** B

31. **Which statement is true about a nested loop?**
    - A) A loop inside another loop
    - B) Two loops running in parallel
    - C) A loop that calls itself
    - D) None of the above
    - **Correct Answer:** A
32. **What is the output of the following code?**

```
cpp
Copy code
int x = 1;
for (int i = 1; i <= 5; i++) {
    x *= i;
}
cout << x;
```

    - A) 5
    - B) 10
    - C) 120
    - D) 1
    - **Correct Answer:** C
33. **What will be the output of this code?**

```
cpp
Copy code
for (int i = 1; i <= 5; i++) {
    if (i == 3) {
        break;
    }
    cout << i << " ";
}
```

    - A) 1 2 3
    - B) 1 2
    - C) 1 2 3 4 5
    - D) 3 4 5
    - **Correct Answer:** B
34. **What is the initial value of `i` in this loop?**

```
cpp
Copy code
for (int i = 1; i < 5; i++) {
}
```

    - A) 0
    - B) 1
    - C) 5
    - D) Undefined
    - **Correct Answer:** B

35. **What will be printed by the following code?**

```
cpp
Copy code
for (int i = 0; i < 3; i++) {
    cout << "Hello ";
}
```

- A) Hello Hello Hello
- B) Hello
- C) Hello Hello
- D) Hello Hello Hello
- **Correct Answer:** D

36. **Which of the following statements correctly terminates a loop?**
    - A) exit
    - B) stop
    - C) break
    - D) return
    - **Correct Answer:** C
37. **What does `cout << i++` output?**
    - A) The current value of i, then increments it
    - B) Increments i first, then outputs the value
    - C) Outputs the incremented value of i
    - D) Outputs nothing
    - **Correct Answer:** A
38. **Which of the following loops will execute at least once?**
    - A) while loop
    - B) for loop
    - C) do-while loop
    - D) All of the above
    - **Correct Answer:** C
39. **How can you print even numbers between 1 and 10 using a loop?**
    - A) Use a for loop with the condition `i % 2 == 0`
    - B) Use a while loop to check `i < 10`
    - C) Use a do-while loop with `i = 0`
    - D) All of the above
    - **Correct Answer:** A
40. **What is the main advantage of using loops in programming?**
    - A) To increase the size of the program
    - B) To reduce redundancy and increase efficiency
    - C) To write more complex code
    - D) To avoid using functions
    - **Correct Answer:** B

## LECTURE 8

## Multiple-Choice Questions (MCQs)

1. **What is the primary purpose of the switch statement in C++?**
   - A) To create loops
   - B) To handle multiple conditions
   - C) To declare variables
   - D) To perform arithmetic operations

   **Answer:** B) To handle multiple conditions
2. **Which statement is used to exit a switch case once a case is matched?**
   - A) continue
   - B) return
   - C) break
   - D) exit

   **Answer:** C) break
3. **What data types can be used with the switch statement?**
   - A) float
   - B) char and integer
   - C) string
   - D) bool

   **Answer:** B) char and integer
4. **In a switch statement, what happens if no case matches the switch expression?**
   - A) The program crashes
   - B) The control jumps to the next case
   - C) The default case is executed if provided
   - D) The switch statement is skipped

   **Answer:** C) The default case is executed if provided
5. **Which of the following statements is TRUE about the break statement?**
   - A) It can only be used in loops
   - B) It ends the program execution
   - C) It prevents fall-through in switch cases
   - D) It must be used after every case in a switch

   **Answer:** C) It prevents fall-through in switch cases
6. **What will happen if you omit the break statement after a case in a switch?**
   - A) The program will not compile
   - B) Only the matching case will execute
   - C) All subsequent cases will execute until a break is encountered
   - D) The default case will execute

   **Answer:** C) All subsequent cases will execute until a break is encountered
7. **Which keyword is used to skip the current iteration of a loop in C++?**
   - A) break
   - B) continue
   - C) skip
   - D) exit

   **Answer:** B) continue
8. **Which of the following is NOT a valid case label in a switch statement?**
   - A) case 1:

- B) case 'A':
- C) case x:
- D) case 0:

**Answer:** C) case x:

9. **If you want to handle both uppercase and lowercase inputs in a switch case, how would you write it?**
   - A) Combine cases with an OR
   - B) Use only one case
   - C) Write separate cases for each
   - D) Use a loop

   **Answer:** C) Write separate cases for each
10. **What type of structure is the switch statement?**
    - A) Sequential
    - B) Conditional
    - C) Iterative
    - D) Recursive

    **Answer:** B) Conditional
11. **What will the following code output if the grade is 'C'?**

```
cpp
Copy code
switch (grade) {
    case 'A': cout << "Excellent";
    case 'B': cout << "Very Good";
    case 'C': cout << "Good";
    case 'D': cout << "Poor";
    case 'F': cout << "Fail";
}
```

    - A) Excellent
    - B) Very Good
    - C) Good
    - D) All of the above

    **Answer:** D) All of the above
12. **What is the purpose of the default case in a switch statement?**
    - A) To execute when no cases match
    - B) To provide an alternative case
    - C) To initialize variables
    - D) To terminate the program

    **Answer:** A) To execute when no cases match
13. **What is an advantage of using a switch statement over multiple if statements?**
    - A) More flexible
    - B) More readable and efficient for multiple conditions
    - C) Easier to debug
    - D) None of the above

    **Answer:** B) More readable and efficient for multiple conditions
14. **In C++, the ASCII value of 'A' is:**
    - A) 97

- B) 65
- C) 32
- D) 100

  **Answer:** B) 65

15. **What will be printed if `grade` is neither 'A', 'B', 'C', 'D', nor 'F', and a default case is provided as `default: cout << "Invalid grade";`?**
    - A) No output
    - B) "Invalid grade"
    - C) A syntax error
    - D) The program will terminate

      **Answer:** B) "Invalid grade"
16. **Which loop allows the use of the break statement to exit immediately?**
    - A) if statement
    - B) for loop
    - C) do-while loop
    - D) All of the above

      **Answer:** D) All of the above
17. **Which of the following best describes the flow of control in a switch statement?**
    - A) Linear
    - B) Random
    - C) Sequential until a break is hit
    - D) Circular

      **Answer:** C) Sequential until a break is hit
18. **What is the result of the following code snippet if `x` equals 10?**

```
cpp
Copy code
switch (x) {
    case 10: cout << "Ten"; break;
    case 20: cout << "Twenty"; break;
    default: cout << "Other";
}
```

    - A) Ten
    - B) Twenty
    - C) Other
    - D) No output

      **Answer:** A) Ten
19. **In which scenario is it appropriate to use the continue statement?**
    - A) When exiting a function
    - B) To skip the rest of the loop's current iteration
    - C) To break out of a loop
    - D) None of the above

      **Answer:** B) To skip the rest of the loop's current iteration
20. **Which of the following is NOT a characteristic of the switch statement?**
    - A) Can handle character data
    - B) Allows compound conditions
    - C) Case labels must be unique within the switch

- D) Control flow can fall through to subsequent cases

  **Answer:** B) Allows compound conditions

## MCQs

1. **What does the switch statement do?**
   - A) Allows multiple conditions to be checked
   - B) Executes only one condition
   - C) Iterates over a collection
   - D) None of the above

     **Answer:** A) Allows multiple conditions to be checked
2. **What is the purpose of the break statement in a switch?**
   - A) To terminate the program
   - B) To exit a loop
   - C) To stop the execution of the current case
   - D) To skip to the next iteration

     **Answer:** C) To stop the execution of the current case
3. **What happens if a break statement is omitted in a switch case?**
   - A) The program stops
   - B) The next case statements will be executed
   - C) The program will generate an error
   - D) The execution will jump to the end of the switch

     **Answer:** B) The next case statements will be executed
4. **In a switch statement, which of the following is NOT a valid case?**
   - A) case 10:
   - B) case 'A':
   - C) case (x + 1):
   - D) case 'a':

     **Answer:** C) case (x + 1):
5. **What is the role of the default statement in a switch?**
   - A) It is mandatory to use
   - B) It executes if no case matches
   - C) It always executes
   - D) None of the above

     **Answer:** B) It executes if no case matches
6. **Which statement allows you to skip the rest of the loop iteration?**
   - A) break
   - B) continue
   - C) exit
   - D) return

     **Answer:** B) continue
7. **In the context of loops, what does the continue statement do?**
   - A) It exits the loop immediately
   - B) It skips the current iteration
   - C) It starts a new loop

- D) None of the above

  **Answer:** B) It skips the current iteration

8. **What is the correct syntax for a switch statement?**
   - A) switch variable { case constant: statement; }
   - B) switch (variable) { case constant: statement; }
   - C) switch (variable) case constant: statement;
   - D) switch variable: case constant: statement;

     **Answer:** B) switch (variable) { case constant: statement; }
9. **Which of the following is a good practice when using a switch statement?**
   - A) Omitting the default case
   - B) Using multiple break statements
   - C) Using goto statements
   - D) Ignoring case sensitivity

     **Answer:** B) Using multiple break statements
10. **What happens if the input grade is 'B' in the following switch case structure?**

```
cpp
Copy code
switch (grade) {
    case 'A': cout << "Excellent";
    case 'B': cout << "Very Good";
    case 'C': cout << "Good";
    default: cout << "Invalid grade";
}
```

    - A) "Very Good" and "Good" will print
    - B) Only "Very Good" will print
    - C) "Excellent" and "Very Good" will print
    - D) The program will crash

      **Answer:** A) "Very Good" and "Good" will print
11. **Which construct allows you to make decisions in programming?**
    - A) Loops
    - B) Functions
    - C) Conditional statements
    - D) Arrays

      **Answer:** C) Conditional statements
12. **When using the continue statement in a while loop, where does the control go next?**
    - A) To the end of the loop
    - B) To the beginning of the loop
    - C) To the next statement after the loop
    - D) It terminates the loop

      **Answer:** B) To the beginning of the loop
13. **Which of the following is true about structured programming?**
    - A) It encourages the use of goto statements
    - B) It promotes the use of sequences, decisions, and loops
    - C) It is less efficient than unstructured programming
    - D) It does not allow functions

      **Answer:** B) It promotes the use of sequences, decisions, and loops

14. **What type of values can be used in the switch statement?**
    - A) Float values
    - B) String values
    - C) Integer and character values
    - D) Any type of variable

    **Answer:** C) Integer and character values
15. **In the provided salary deduction example, what deduction is made if the salary is 15,000?**
    - A) 0
    - B) 1,000
    - C) 7%
    - D) 1,500

    **Answer:** B) 1,000
16. **What is the primary purpose of using the break statement?**
    - A) To continue the loop
    - B) To exit the current case in switch
    - C) To end the program
    - D) To skip a line of code

    **Answer:** B) To exit the current case in switch
17. **What happens when the continue statement is executed in a for loop?**
    - A) The loop terminates
    - B) The loop restarts from the next iteration
    - C) The program crashes
    - D) None of the above

    **Answer:** B) The loop restarts from the next iteration
18. **Why is it advised to avoid using goto statements?**
    - A) They are not allowed in C++
    - B) They make code harder to read and maintain
    - C) They are inefficient
    - D) They cause compilation errors

    **Answer:** B) They make code harder to read and maintain
19. **How should the input grade be handled to accommodate both uppercase and lowercase?**
    - A) Use only uppercase letters
    - B) Use only lowercase letters
    - C) Use separate cases for both
    - D) Ignore case sensitivity

    **Answer:** C) Use separate cases for both
20. **What should be included in every switch statement for better practice?**
    - A) A default case
    - B) An exit statement
    - C) A loop
    - D) A global variable

    **Answer:** A) A default case

## LECTURE 9

## Multiple Choice Questions (MCQs)

1. **What is a function in programming?**
   - A) A method to store data
   - B) A reusable block of code that performs a specific task
   - C) A variable that changes value
   - D) An error in code

   **Answer: B**
2. **What does the term 'top-down design' refer to?**
   - A) Writing code from bottom to top
   - B) Breaking a large problem into smaller sub-tasks
   - C) Creating a main function first
   - D) Compiling code

   **Answer: B**
3. **Which of the following is NOT a type of function?**
   - A) Functions that return a value
   - B) Functions that do not return a value
   - C) Functions that perform I/O operations
   - D) Functions that only perform calculations

   **Answer: D**
4. **In C++, what keyword is used to return a value from a function?**
   - A) exit
   - B) end
   - C) return
   - D) finish

   **Answer: C**
5. **What is the default return type of a function if not specified?**
   - A) void
   - B) int
   - C) double
   - D) char

   **Answer: A**
6. **What type of argument can be passed to a function?**
   - A) Only integers
   - B) Only floats
   - C) Any valid data type
   - D) Only characters

   **Answer: C**
7. **What is the purpose of function declaration?**
   - A) To execute the function
   - B) To inform the compiler about the function's existence
   - C) To define the function
   - D) To create a loop

   **Answer: B**
8. **Which of the following correctly declares a function?**
   - A) int functionName(int);

- B) functionName(int) int;
- C) functionName(int) = int;
- D) void functionName;

**Answer: A**

9. **What does the argument list in a function definition specify?**
   - A) The variables to be used
   - B) The data types of parameters the function will take
   - C) The return type of the function
   - D) The scope of the function

   **Answer: B**
10. **Which of the following is an example of call by value?**
    - A) Passing the address of a variable
    - B) Passing a copy of the variable's value
    - C) Using global variables
    - D) Modifying the variable in the calling function

    **Answer: B**
11. **In C++, how are functions called?**
    - A) Using their name followed by parentheses
    - B) By their return type
    - C) By declaring them as global
    - D) By defining them at the end of the program

    **Answer: A**
12. **What is the return type of a function that does not return any value?**
    - A) int
    - B) void
    - C) float
    - D) char

    **Answer: B**
13. **Which symbol is used to define the beginning and end of a function in C++?**
    - A) {}
    - B) ()
    - C) []
    - D) <>

    **Answer: A**
14. **What will happen if a function is called without being declared first?**
    - A) The program will compile without errors
    - B) The program will produce a runtime error
    - C) The compiler will give an error
    - D) The function will not execute

    **Answer: C**
15. **What is the result of the following function call: `square(5)` if `square` is defined to return `number * number`?**
    - A) 5
    - B) 10
    - C) 25

- D) 30

**Answer: C**

16. **What type of function is `int isEven(int)`?**
    - A) It returns a boolean value
    - B) It returns an integer value
    - C) It does not return a value
    - D) It returns a float

    **Answer: B**
17. **In a function definition, which part specifies what the function does?**
    - A) Function declaration
    - B) Function header
    - C) Function body
    - D) Function call

    **Answer: C**
18. **Which of the following correctly defines a function to calculate the square of a number?**
    - A) int square(int number) { return number * number; }
    - B) int square(number) { return number * number; }
    - C) void square(int number) { return number * number; }
    - D) int square(number) { return number * number; }

    **Answer: A**
19. **What keyword do you use to declare a function that takes no arguments?**
    - A) void
    - B) null
    - C) empty
    - D) none

    **Answer: A**
20. **If a function is defined after the `main()` function, what must be done?**
    - A) Nothing, it works fine
    - B) Declare the function before `main()`
    - C) Move the `main()` function to the end
    - D) Compile it separately

    **Answer: B**
21. **What type of argument should the function `double raiseToPow(double x, int power)` accept for `x`?**
    - A) int
    - B) float
    - C) double
    - D) char

    **Answer: C**
22. **How can you identify a function that returns a boolean value?**
    - A) By its name
    - B) By its return type
    - C) By its arguments
    - D) By its body

    **Answer: B**

23. **Which of the following is an example of using a function as an expression?**
    - A) square(5);
    - B) int result = square(5);
    - C) void square(int);
    - D) all of the above

    **Answer: B**
24. **In the function `double circleArea(double radius)`, what does `radius` represent?**
    - A) The return value
    - B) A local variable
    - C) A parameter
    - D) A global variable

    **Answer: C**
25. **What is the result of the following statement: `result = square(5) + square(3)` if `square` returns `number * number`?**
    - A) 8
    - B) 25
    - C) 34
    - D) 28

    **Answer: C**
26. **If a function modifies the value of the argument passed to it, what kind of passing mechanism is used?**
    - A) Call by value
    - B) Call by reference
    - C) Global passing
    - D) Local passing

    **Answer: B**
27. **What is an important benefit of using functions in programming?**
    - A) They increase code complexity
    - B) They reduce code reuse
    - C) They provide modularity
    - D) They eliminate the need for comments

    **Answer: C**
28. **Which function call correctly invokes a function that takes two integers?**
    - A) functionName(10, 20);
    - B) functionName(int x, int y);
    - C) functionName(x, y);
    - D) functionName(10);

    **Answer: A**
29. **How do you specify a function that takes a single integer argument and returns nothing?**
    - A) void functionName(int);
    - B) int functionName(void);
    - C) functionName(int);
    - D) void functionName;

    **Answer: A**
30. **What will the following code output?**

```
cpp
Copy code
int main() {
    int x = 3;
    cout << square(x + 2);
}
```

- A) 5
- B) 25
- C) 9
- D) 6

**Answer: B**

31. **What keyword is used to indicate a function will not return any value?**
    - A) return
    - B) void
    - C) empty
    - D) none

    **Answer: B**
32. **Which statement about function overloading is true?**
    - A) Functions must have different names
    - B) Functions must have the same return type
    - C) Functions must have different parameter lists
    - D) Functions must be in separate files

    **Answer: C**
33. **What does the term 'encapsulation' mean in the context of functions?**
    - A) Hiding the function code from the caller
    - B) Making the function accessible globally
    - C) Combining multiple functions into one
    - D) Creating a separate header file

    **Answer: A**
34. **What is the correct way to define a function prototype?**
    - A) functionName(int, float);
    - B) void functionName(int, float);
    - C) functionName(int x, float y);
    - D) void functionName;

    **Answer: B**
35. **If a function `multiply` takes two parameters and returns their product, how would you call it?**
    - A) multiply(5);
    - B) multiply(5, 10);
    - C) multiply(int, int);
    - D) multiply(5, 10, 15);

    **Answer: B**
36. **What happens if a function is declared but not defined?**
    - A) The program will run without errors
    - B) The program will not compile
    - C) The function can still be called

- D) The function executes but does nothing

  **Answer: B**

37. **How can functions help in debugging code?**
    - A) They can make code more complex
    - B) They help isolate errors in smaller code blocks
    - C) They remove the need for debugging
    - D) They automatically fix errors

      **Answer: B**
38. **Which of the following is a valid function signature?**
    - A) void 1function();
    - B) void function_1();
    - C) void function name();
    - D) void function-name();

      **Answer: B**
39. **Which of the following statements is TRUE about local variables in functions?**
    - A) They can be accessed from anywhere in the program
    - B) They exist only within the function
    - C) They retain their values between function calls
    - D) They cannot be initialized

      **Answer: B**
40. **What is the primary advantage of using default arguments in functions?**
    - A) They make functions less flexible
    - B) They allow functions to be called with fewer arguments
    - C) They increase the number of function definitions needed
    - D) They eliminate the need for function declarations

      **Answer: B**

# LECTURE 10

## MCQs on C++ Programming

1. **What is a header file in C++?**
    - A) A file containing implementation of functions
    - B) A file containing function prototypes and constants
    - C) A file for input/output operations
    - D) A file for error handling

      **Answer:** B
2. **Which directive is used to include a header file?**
    - A) #include
    - B) #define
    - C) #includeFile
    - D) #import

      **Answer:** A
3. **What is a function prototype?**
    - A) The definition of a function

- B) A declaration of a function that specifies its return type and parameters
- C) The implementation of a function
- D) A variable declaration

  **Answer:** B

4. **In C++, what does the keyword `void` indicate in a function declaration?**
   - A) The function does not take parameters
   - B) The function does not return a value
   - C) The function returns an integer
   - D) The function is private

     **Answer:** B
5. **What is the scope of a global variable?**
   - A) Only within the function
   - B) Only within a code block
   - C) Throughout the entire program
   - D) Only within a class

     **Answer:** C
6. **What will happen if two variables with the same name are declared in different scopes?**
   - A) Compilation error
   - B) The program will use the first declared variable
   - C) The inner variable will override the outer variable
   - D) They are treated as one variable

     **Answer:** C
7. **What does "call by value" mean?**
   - A) Passing the address of a variable to a function
   - B) Passing a reference of a variable to a function
   - C) Passing a copy of the variable's value to a function
   - D) Returning a value from a function

     **Answer:** C
8. **How do you pass a variable by reference in C++?**
   - A) By using the `*` operator
   - B) By using the `&` operator
   - C) By using the `#` operator
   - D) By using the `@` operator

     **Answer:** B
9. **Which of the following is a valid function declaration?**
   - A) `int function(int x);`
   - B) `function int(int x);`
   - C) `int function(int x)`
   - D) `int function;`

     **Answer:** A
10. **What will the following code print?**

```
cpp
Copy code
int x = 5;
cout << x++;
```

- A) 5
- B) 6
- C) Compilation error
- D) 0

**Answer:** A

11. **What is the purpose of the `#define` directive?**
    - A) To include libraries
    - B) To define constants
    - C) To declare functions
    - D) To manage memory

    **Answer:** B
12. **What will be the output of the following code?**

```
cpp
Copy code
void func() {
    int a = 10;
    cout << a;
}
func();
```

    - A) 10
    - B) Compilation error
    - C) 0
    - D) Undefined

    **Answer:** A
13. **What type of function can call itself?**
    - A) Recursive function
    - B) Iterative function
    - C) Overloaded function
    - D) Inline function

    **Answer:** A
14. **What does the following function return?**

```
cpp
Copy code
long fact(long n) {
    if (n <= 1)
        return 1;
    else
        return n * fact(n - 1);
}
```

    - A) The square of `n`
    - B) The factorial of `n`
    - C) The sum of integers up to `n`
    - D) None of the above

    **Answer:** B
15. **In the context of C++, what is a local variable?**

- A) A variable declared inside a function or block
- B) A variable that can be accessed globally
- C) A constant value
- D) A static variable

**Answer:** A

16. **What happens if you do not provide a return type for a function?**
    - A) Compilation error
    - B) The function returns `void` by default
    - C) The function returns `int` by default
    - D) The function does not execute

    **Answer:** A
17. **Which of the following will cause a compilation error?**
    - A) `int x; x = 5;`
    - B) `double y; y = 5.0;`
    - C) `int z; z = "Hello";`
    - D) `char c = 'A';`

    **Answer:** C
18. **Which operator is used to obtain the address of a variable?**
    - A) *
    - B) &
    - C) %
    - D) @

    **Answer:** B
19. **What is the result of the following code?**

```
cpp
Copy code
int a = 5;
int b = 10;
cout << a + b;
```

    - A) 15
    - B) 510
    - C) Compilation error
    - D) 5

    **Answer:** A
20. **Which statement is true regarding call by reference?**
    - A) It is safer than call by value
    - B) It requires more memory
    - C) It allows modification of the original variable
    - D) It is slower than call by value

    **Answer:** C
21. **Which of the following is NOT a valid identifier in C++?**
    - A) myVariable
    - B) 2ndVariable
    - C) variable_1

- D) variable$

**Answer:** B

22. **What is the correct way to declare a constant in C++?**
    - A) const int x = 10;
    - B) constant int x = 10;
    - C) #define x 10
    - D) int x = 10 const;

    **Answer:** A
23. **What will be the output of the following code?**

```
cpp
Copy code
int x = 5;
cout << (x > 2 ? "Greater" : "Smaller");
```

    - A) Greater
    - B) Smaller
    - C) 5
    - D) Compilation error

    **Answer:** A
24. **What type of loop is NOT supported in C++?**
    - A) for loop
    - B) while loop
    - C) do-while loop
    - D) repeat-until loop

    **Answer:** D
25. **Which of the following is a valid comment in C++?**
    - A) // This is a comment
    - B) /* This is a comment */
    - C) Both A and B
    - D) None of the above

    **Answer:** C
26. **What will the following code output?**

```
cpp
Copy code
int main() {
    int a = 3, b = 4;
    cout << a * b;
}
```

    - A) 12
    - B) 7
    - C) 34
    - D) Compilation error

    **Answer:** A
27. **Which function is used to get input from the user in C++?**
    - A) cin

- B) input
- C) read
- D) get

**Answer:** A

28. **What does the `return` statement do in a function?**
    - A) Stops the function execution
    - B) Returns a value to the calling function
    - C) Both A and B
    - D) None of the above

    **Answer:** C
29. **What will happen if a function is called with fewer arguments than it expects?**
    - A) Compilation error
    - B) The program will crash
    - C) Default values will be assigned
    - D) The function will execute with garbage values

    **Answer:** A
30. **Which of the following is a valid way to define an array in C++?**
    - A) int arr[10];
    - B) int arr = new int[10];
    - C) int arr(10);
    - D) None of the above

    **Answer:** A
31. **What is the output of the following code?**

```
cpp
Copy code
for(int i = 0; i < 5; i++) {
    cout << i << " ";
}
```

    - A) 1 2 3 4 5
    - B) 0 1 2 3 4
    - C) 0 1 2 3 4 5
    - D) 1 2 3 4

    **Answer:** B
32. **Which statement is true regarding arrays?**
    - A) Arrays can change size dynamically
    - B) All elements in an array must be of the same type
    - C) Arrays can only hold integers
    - D) Arrays can be of any data type

    **Answer:** B
33. **What is the default return type of a function in C++ if none is specified?**
    - A) int
    - B) void
    - C) double
    - D) float

    **Answer:** A

34. **What is the purpose of the `break` statement?**
    - A) To exit a loop
    - B) To stop the program
    - C) To return from a function
    - D) To define a constant

    **Answer:** A
35. **What will the following code print?**

```
cpp
Copy code
for(int i = 0; i < 3; i++) {
    for(int j = 0; j < 2; j++) {
        cout << i << j << " ";
    }
}
```

    - A) 012 013
    - B) 00 01 10 11
    - C) 00 01 10 12 20 21
    - D) 000 001 010 011

    **Answer:** C
36. **Which operator is used to access members of a structure?**
    - A) .
    - B) ->
    - C) ::
    - D) &

    **Answer:** A
37. **What will happen if you use the delete operator on a non-dynamically allocated variable?**
    - A) Compilation error
    - B) Undefined behavior
    - C) The variable will be deleted
    - D) None of the above

    **Answer:** B
38. **Which of the following is a type of constructor?**
    - A) Default constructor
    - B) Parameterized constructor
    - C) Copy constructor
    - D) All of the above

    **Answer:** D
39. **What is the purpose of the `static` keyword?**
    - A) To declare a global variable
    - B) To create a constant variable
    - C) To limit the visibility of a variable
    - D) To declare a variable that retains its value between function calls

    **Answer:** D
40. **What is the primary purpose of a destructor in C++?**
    - A) To initialize a class

- B) To allocate memory
- C) To clean up resources when an object is destroyed
- D) To modify the object's state

  **Answer:** C

# LECTURE 11

## MCQs on Arrays and Linear Search

1. **What is the correct syntax to declare an array of integers in C++?**
   - A) `int array[10];`
   - B) `array int[10];`
   - C) `int[10] array;`
   - D) `declare int array[10];`
   - **Answer:** A
2. **What does the index of an array start from in C++?**
   - A) 1
   - B) 0
   - C) -1
   - D) Depends on the declaration
   - **Answer:** B
3. **How many elements does the array `int ages[5];` hold?**
   - A) 4
   - B) 5
   - C) 6
   - D) 10
   - **Answer:** B
4. **Which of the following correctly initializes an array of integers to zero?**
   - A) `int arr[5] = {0};`
   - B) `int arr[5]; arr = {0};`
   - C) `int arr[5] = {1, 1, 1, 1, 1};`
   - D) `int arr[5];`
   - **Answer:** A
5. **Which loop is ideal for iterating through all elements of an array?**
   - A) do-while loop
   - B) for loop
   - C) while loop
   - D) None of the above
   - **Answer:** B
6. **What is the maximum index of the array `int nums[10];`?**
   - A) 9
   - B) 10
   - C) 11
   - D) 0
   - **Answer:** A

7. **Which of the following statements correctly copies elements from one array to another?**
   - A) `b = a;`
   - B) `for(i = 0; i < 10; i++) b[i] = a[i];`
   - C) `b.copy(a);`
   - D) `b = copy(a);`
   - **Answer:** B
8. **What does the keyword `const` indicate when used with an array size?**
   - A) The array is immutable.
   - B) The array size cannot be changed.
   - C) The array can store constant values only.
   - D) None of the above.
   - **Answer:** B
9. **In a linear search, what is the worst-case time complexity?**
   - A) O(1)
   - B) O(n)
   - C) O(log n)
   - D) $O(n^2)$
   - **Answer:** B
10. **Which statement is true about array elements in C++?**
    - A) They can be of different data types.
    - B) They must be of the same data type.
    - C) They are always initialized to zero.
    - D) None of the above.
    - **Answer:** B
11. **What is the purpose of the `break` statement in a loop?**
    - A) To continue the loop.
    - B) To terminate the loop immediately.
    - C) To skip the current iteration.
    - D) To reset the loop counter.
    - **Answer:** B
12. **How can you prevent array out-of-bounds errors?**
    - A) Use a loop to check bounds.
    - B) Always initialize arrays.
    - C) Declare larger arrays.
    - D) Use pointers instead of arrays.
    - **Answer:** A
13. **Which of the following correctly declares an array of size 100?**
    - A) `int arr(100);`
    - B) `int arr[100];`
    - C) `int arr;`
    - D) `int[100] arr;`
    - **Answer:** B
14. **If you want to read user input into an array of size 10, which code snippet is correct?**
    - A) `for(int i = 0; i < 10; i++) cin >> arr[i];`

- B) `cin >> arr[10];`
- C) `for(int i = 1; i <= 10; i++) cin >> arr[i];`
- D) `int arr[10]; arr = input();`
- **Answer:** A

15. **What is the effect of this code: `int arr[5] = {1, 2, 3};`?**
    - A) Only the first three elements are initialized; others are garbage.
    - B) All elements are initialized to 0.
    - C) Compilation error.
    - D) The array will have elements 1, 2, 3, 0, 0.
    - **Answer:** D
16. **Which operator can be used to find the remainder of a division?**
    - A) /
    - B) *
    - C) %
    - D) ^
    - **Answer:** C
17. **What will happen if you access an array element using an index greater than its size?**
    - A) The program will crash.
    - B) The value will be zero.
    - C) It will access the next memory location.
    - D) It will return a compilation error.
    - **Answer:** C
18. **Which of the following is true regarding the `rand()` function?**
    - A) It generates a random floating-point number.
    - B) It can generate negative numbers.
    - C) It requires the `<stdlib.h>` header.
    - D) It generates numbers only between 0 and 10.
    - **Answer:** C
19. **What does the line `int ages[10] = {1, 2, 3, 4, 5};` do?**
    - A) Initializes all elements to 1.
    - B) Initializes first five elements; the rest are set to 0.
    - C) Causes a compilation error.
    - D) Initializes all elements to 5.
    - **Answer:** B
20. **What is the main benefit of using arrays?**
    - A) They can store different types of data.
    - B) They allow for efficient data management of similar items.
    - C) They automatically expand in size.
    - D) They are easier to debug.
    - **Answer:** B
21. **In a linear search algorithm, what happens if the number is not found?**
    - A) The algorithm will throw an exception.
    - B) It will return the index -1.
    - C) It will run indefinitely.
    - D) The algorithm will terminate successfully.

- **Answer:** B

22. **What is a common method to initialize an array in C++?**
    - A) Using the assignment operator.
    - B) Using a loop.
    - C) Both A and B.
    - D) None of the above.
    - **Answer:** C
23. **How do you access the first element of an array named `arr`?**
    - A) `arr[0]`
    - B) `arr[1]`
    - C) `arr.first`
    - D) `arr(0)`
    - **Answer:** A
24. **Which of the following is NOT a valid array initialization in C++?**
    - A) `int arr[3] = {1, 2};`
    - B) `int arr[5] = {1, 2, 3, 4, 5};`
    - C) `int arr[] = {1, 2, 3};`
    - D) `int arr[3] = {1, 2, 3, 4};`
    - **Answer:** D
25. **In the context of arrays, what does "contiguous memory" mean?**
    - A) Memory is allocated in different segments.
    - B) Memory addresses are sequentially allocated.
    - C) Memory allocation is dynamic.
    - D) None of the above.
    - **Answer:** B
26. **What will the following code snippet print?**

```
cpp
Copy code
int arr[3] = {10, 20, 30};
cout << arr[1];
```

    - A) 10
    - B) 20
    - C) 30
    - D) Undefined behavior
    - **Answer:** B
27. **Which of the following best describes a "for loop"?**
    - A) It iterates at least once.
    - B) It is used for conditional loops.
    - C) It iterates a specific number of times.
    - D) It runs indefinitely.
    - **Answer:** C
28. **In a linear search, if the search is successful, what does the algorithm return?**
    - A) The element found.
    - B) The index of the element.
    - C) A boolean value.

- D) The size of the array.
- **Answer:** B

29. **Which statement about arrays is false?**
    - A) Arrays can be of any data type.
    - B) Array size must be a constant expression.
    - C) Arrays can store objects of different classes.
    - D) Arrays are zero-indexed.
    - **Answer:** C
30. **How do you declare a constant size array?**
    - A) `const int size = 10; int arr[size];`
    - B) `int const arr[10];`
    - C) `const int arr[10];`
    - D) `const int arr[] = {10};`
    - **Answer:** C
31. **What will this code output?**

```
cpp
Copy code
int arr[5] = {0};
cout << arr[4];
```

    - A) 0
    - B) 1
    - C) Undefined
    - D) Garbage value
    - **Answer:** A
32. **How do you find the length of an array in C++?**
    - A) Using `sizeof(array);`
    - B) There is no built-in function.
    - C) `length(array);`
    - D) By iterating until a NULL element.
    - **Answer:** A
33. **In a linear search, if the target value is found at index 4, what will be the output of `found = true; break;`?**
    - A) Continue searching.
    - B) Terminate the search and set found to true.
    - C) Set found to false.
    - D) None of the above.
    - **Answer:** B
34. **Which function can be used to generate a random number in a specified range?**
    - A) `rand(int, int)`
    - B) `rand() % n`
    - C) `randomize()`
    - D) `rand(1, 6)`
    - **Answer:** B
35. **What will this code snippet output?**

```
cpp
Copy code
int arr[3] = {5, 10, 15};
cout << arr[3];
```

- A) 5
- B) 10
- C) 15
- D) Undefined behavior
- **Answer:** D

36. **What type of loop is typically used to copy array elements?**
    - A) for loop
    - B) do-while loop
    - C) infinite loop
    - D) nested loop
    - **Answer:** A
37. **How do you safely iterate over an array in C++?**
    - A) Use an index without bounds checking.
    - B) Always check the index against the array size.
    - C) Use `foreach` syntax.
    - D) Iteration is not safe in C++.
    - **Answer:** B
38. **Which statement about a linear search is true?**
    - A) It is the fastest search algorithm.
    - B) It works only on sorted arrays.
    - C) It checks each element sequentially.
    - D) It uses binary search technique.
    - **Answer:** C
39. **What is the value of `arr[2]` after this initialization: `int arr[5] = {1, 2};`?**
    - A) 1
    - B) 2
    - C) 0
    - D) Undefined
    - **Answer:** C
40. **What will happen if you try to initialize an array with more elements than its declared size?**
    - A) It will compile successfully.
    - B) It will throw a runtime error.
    - C) It will result in a compilation error.
    - D) It will initialize only the first few elements.
    - **Answer:** C

# LECTURE 12

## MCQs on Arrays in C++

1. **What is a character array used for in C++?**
   - A) Storing integers
   - B) Storing strings
   - C) Storing floating-point numbers
   - D) None of the above

   **Answer:** B) Storing strings
2. **How is a string in a character array terminated in C++?**
   - A) With a space
   - B) With a semicolon
   - C) With a null character
   - D) With a comma

   **Answer:** C) With a null character
3. **What is the correct way to declare a character array to hold a name in C++?**
   - A) `char name[100];`
   - B) `char name(100);`
   - C) `char name{100};`
   - D) `string name;`

   **Answer:** A) `char name[100];`
4. **Which of the following correctly initializes a character array?**
   - A) `char name[100] = {'H', 'e', 'l', 'l', 'o'};`
   - B) `char name[100] = "Hello";`
   - C) `char name = "Hello";`
   - D) Both A and B

   **Answer:** D) Both A and B
5. **What does the `cin` statement do when reading a string into a character array?**
   - A) Reads until a space is encountered
   - B) Reads the entire line including spaces
   - C) Reads until the enter key is pressed
   - D) Ignores all inputs

   **Answer:** C) Reads until the enter key is pressed
6. **In a character array, what character marks the end of the string?**
   - A) `*`
   - B) `#`
   - C) `\0`
   - D) `@`

   **Answer:** C) `\0`
7. **How do you compare two character arrays in C++?**
   - A) Using the == operator
   - B) Using a loop to compare each character
   - C) Using the `strcmp` function
   - D) Both B and C

   **Answer:** D) Both B and C
8. **What is the purpose of the sorting algorithm for arrays?**
   - A) To arrange elements in random order
   - B) To store elements in reverse order
   - C) To arrange elements in ascending or descending order

- D) To delete elements

  **Answer:** C) To arrange elements in ascending or descending order

9. **What is the main advantage of using a binary search over a linear search?**
   - A) Binary search can find all duplicates
   - B) Binary search is faster with sorted arrays
   - C) Binary search works with unsorted arrays
   - D) None of the above

   **Answer:** B) Binary search is faster with sorted arrays
10. **What happens when you pass an array to a function in C++?**
    - A) A copy of the array is made
    - B) The array is passed by reference
    - C) The array is passed by value
    - D) The function cannot access the array

    **Answer:** B) The array is passed by reference
11. **What is the output of `cout << name;` if `name` contains "Alice"?**
    - A) Alice
    - B) "Alice"
    - C) The memory address of `name`
    - D) None of the above

    **Answer:** A) Alice
12. **What is the size of an array declared as `int arr[10];`?**
    - A) 9
    - B) 10
    - C) 11
    - D) 0

    **Answer:** B) 10
13. **How can you access the third element of an integer array `arr`?**
    - A) `arr[3]`
    - B) `arr(3)`
    - C) `arr[2]`
    - D) `arr{2}`

    **Answer:** C) `arr[2]`
14. **In C++, what is the result of `if (a != b)` when `a` and `b` are equal?**
    - A) True
    - B) False
    - C) Undefined
    - D) It causes an error

    **Answer:** B) False
15. **Which of the following is NOT a characteristic of arrays in C++?**
    - A) Fixed size
    - B) Can hold multiple data types
    - C) Indexed access
    - D) Contiguous memory allocation

    **Answer:** B) Can hold multiple data types
16. **Which sorting algorithm repeatedly selects the smallest element from an unsorted list?**

- A) Insertion Sort
- B) Selection Sort
- C) Bubble Sort
- D) Quick Sort

  **Answer:** B) Selection Sort

17. **When searching for an element in a sorted array, what does binary search do first?**
    - A) Compares the element with the last element
    - B) Finds the middle element and compares
    - C) Looks at the first element only
    - D) Counts the number of elements

      **Answer:** B) Finds the middle element and compares
18. **Which function is used in C++ to find the length of a string stored in a character array?**
    - A) `length()`
    - B) `size()`
    - C) `strlen()`
    - D) `count()`

      **Answer:** C) `strlen()`
19. **What happens if you try to access an index outside the bounds of an array?**
    - A) The program crashes
    - B) You get garbage values
    - C) It causes a syntax error
    - D) None of the above

      **Answer:** B) You get garbage values
20. **Which of the following correctly implements a linear search?**
    - A) A loop that checks each element one by one
    - B) A recursive function
    - C) Sorting the array first
    - D) Using a built-in function

      **Answer:** A) A loop that checks each element one by one

## Multiple-Choice Questions (MCQs)

1. **What is the index of the first element in a C++ array?**
    - A) 0
    - B) 1
    - C) -1
    - D) None of the above

      **Answer:** A) 0
2. **How do you declare a one-dimensional array of integers with 10 elements?**
    - A) `int array(10);`
    - B) `int array[10];`
    - C) `array int[10];`
    - D) `int[10] array;`

      **Answer:** B) `int array[10];`
3. **What is the output of the following code?**

```
cpp
Copy code
int arr[3] = {1, 2, 3};
cout << arr[1];
```

- A) 1
- B) 2
- C) 3
- D) Error

**Answer:** B) 2

4. **Which of the following correctly initializes a two-dimensional array of integers with 2 rows and 3 columns?**
   - A) `int matrix[2][3];`
   - B) `int[2][3] matrix;`
   - C) `int matrix(2, 3);`
   - D) `int matrix{2, 3};`

   **Answer:** A) `int matrix[2][3];`
5. **How can you access the third element of an array named `myArray`?**
   - A) `myArray[2]`
   - B) `myArray[3]`
   - C) `myArray(2)`
   - D) `myArray[1]`

   **Answer:** A) `myArray[2]`
6. **What will be the output of the following code?**

```
cpp
Copy code
int arr[5] = {0, 1, 2, 3, 4};
cout << arr[4];
```

   - A) 3
   - B) 4
   - C) 5
   - D) Error

   **Answer:** B) 4
7. **Which operator is used to get the size of an array in C++?**
   - A) `sizeof()`
   - B) `length()`
   - C) `size()`
   - D) `count()`

   **Answer:** A) `sizeof()`
8. **What happens if you try to access an out-of-bounds index of an array?**
   - A) It gives a compilation error
   - B) It returns a default value
   - C) It leads to undefined behavior
   - D) It crashes the program

   **Answer:** C) It leads to undefined behavior

9. **What type of loop is commonly used to iterate over an array?**
    - A) For loop
    - B) While loop
    - C) Do-while loop
    - D) All of the above

    **Answer:** D) All of the above
10. **Which of the following is a valid way to declare an array in C++?**
    - A) `int arr[] = {1, 2, 3};`
    - B) `int arr[3] = 1, 2, 3;`
    - C) `int arr(3) = {1, 2, 3};`
    - D) `int arr = {1, 2, 3};`

    **Answer:** A) `int arr[] = {1, 2, 3};`
11. **Which of the following can be used to pass an array to a function?**
    - A) By value
    - B) By reference
    - C) Both A and B
    - D) None of the above

    **Answer:** B) By reference
12. **What is the output of the following code?**

```
cpp
Copy code
int arr[3] = {1, 2, 3};
arr[0] = arr[1] + arr[2];
cout << arr[0];
```

    - A) 2
    - B) 3
    - C) 5
    - D) 6

    **Answer:** C) 5
13. **Which of the following statements correctly creates a 3-dimensional array?**
    - A) `int arr[3][4][5];`
    - B) `int arr(3, 4, 5);`
    - C) `int arr{3, 4, 5};`
    - D) `int arr[3, 4, 5];`

    **Answer:** A) `int arr[3][4][5];`
14. **How do you declare an array of 5 floating-point numbers?**
    - A) `float arr[5];`
    - B) `float arr(5);`
    - C) `float arr{5};`
    - D) `arr[5] float;`

    **Answer:** A) `float arr[5];`
15. **Which keyword is used to define a constant array in C++?**
    - A) `const`
    - B) `static`
    - C) `final`

- D) `immutable`

**Answer:** A) `const`

16. **What does the following code print?**

```
cpp
Copy code
int arr[3] = {1, 2, 3};
cout << arr[1] + arr[2];
```

- A) 4
- B) 5
- C) 6
- D) 3

**Answer:** B) 5

17. **In a two-dimensional array, how do you access the element in the first row and second column?**
    - A) `array[1][2]`
    - B) `array[0][1]`
    - C) `array(1, 2)`
    - D) `array[0, 1]`

    **Answer:** B) `array[0][1]`
18. **What will the following code output?**

```
cpp
Copy code
int arr[5] = {1, 2, 3};
cout << arr[3];
```

- A) 0
- B) 3
- C) Undefined
- D) Error

**Answer:** A) 0

19. **Which function can be used to find the length of a C-style string stored in a character array?**
    - A) `size()`
    - B) `length()`
    - C) `strlen()`
    - D) `count()`

    **Answer:** C) `strlen()`
20. **How do you find the total number of elements in an array?**
    - A) `sizeof(array)/sizeof(array[0])`
    - B) `length(array)`
    - C) `count(array)`
    - D) `sizeof(array)`

    **Answer:** A) `sizeof(array)/sizeof(array[0])`
21. **What is the result of the following operation?**

```
cpp
Copy code
int arr[4] = {1, 2, 3, 4};
cout << arr[2] * 2;
```

- A) 2
- B) 4
- C) 6
- D) 8

**Answer:** D) 6

22. **Which of the following statements will correctly copy the contents of one array to another?**
    - A) `array2 = array1;`
    - B) `memcpy(array2, array1, sizeof(array1));`
    - C) `copy(array1, array2);`
    - D) None of the above

    **Answer:** B) `memcpy(array2, array1, sizeof(array1));`
23. **What is the default value of uninitialized static array elements in C++?**
    - A) 0
    - B) Garbage value
    - C) 1
    - D) None of the above

    **Answer:** A) 0
24. **In C++, arrays are stored in which type of memory?**
    - A) Stack
    - B) Heap
    - C) Static
    - D) All of the above

    **Answer:** D) All of the above
25. **What happens if you initialize an array with more elements than its size?**
    - A) Compiler error
    - B) Runtime error
    - C) Extra elements are ignored
    - D) Undefined behavior

    **Answer:** A) Compiler error

# LECTURE 13

## MCQs on Array Manipulation in C++

1. **What is an array?**
    - A) A collection of different data types
    - B) A collection of identical or similar data types
    - C) A collection of functions
    - D) A data structure that can hold only integers

    **Correct Answer:** B

2. **In a two-dimensional array, how are elements accessed?**
    - A) Using a single index
    - B) Using two indices
    - C) Using three indices
    - D) Using a pointer

    **Correct Answer:** B
3. **What does the following code snippet declare?** `int a[3][3];`
    - A) A one-dimensional array
    - B) A two-dimensional array with 3 rows and 3 columns
    - C) A pointer to an array
    - D) A single integer

    **Correct Answer:** B
4. **Which loop structure is typically used to process a two-dimensional array?**
    - A) Single loop
    - B) Nested loops
    - C) Recursive function
    - D) Conditional statements

    **Correct Answer:** B
5. **What is the purpose of the `cout` statement in the array input loop?**
    - A) To read values from the array
    - B) To display the values in the array
    - C) To prompt the user for input
    - D) To declare the array

    **Correct Answer:** C
6. **What does the term "flipping the matrix" refer to?**
    - A) Changing the data type of the matrix
    - B) Reversing the order of rows in the matrix
    - C) Transposing the matrix
    - D) Multiplying the matrix by a scalar

    **Correct Answer:** B
7. **In the context of arrays, what does `maxRows` represent?**
    - A) The maximum number of columns
    - B) The maximum number of rows
    - C) The total number of elements
    - D) The maximum value an element can hold

    **Correct Answer:** B
8. **Which character is used to insert a tab in output?**
    - A) `\n`
    - B) `\t`
    - C) `\r`
    - D) `\s`

    **Correct Answer:** B
9. **What will happen if you attempt to access an array element with an index out of bounds?**
    - A) It will return 0
    - B) It will return a random value

- C) It will cause a compile-time error
- D) It may cause a runtime error

  **Correct Answer:** D

10. **What is the main benefit of using arrays?**
    - A) They can store different data types
    - B) They can be resized dynamically
    - C) They allow efficient storage and retrieval of similar data types
    - D) They are easier to debug

      **Correct Answer:** C
11. **What does the flowchart generally represent in programming?**
    - A) The syntax of the programming language
    - B) The logic and flow of the program
    - C) The graphical user interface
    - D) The error handling mechanism

      **Correct Answer:** B
12. **In the nested loop structure for inputting a matrix, what does the outer loop control?**
    - A) The columns of the matrix
    - B) The individual elements
    - C) The rows of the matrix
    - D) The number of iterations

      **Correct Answer:** C
13. **When displaying a flipped matrix, which loop order is used?**
    - A) From the first row to the last row
    - B) From the last row to the first row
    - C) Random order
    - D) No specific order

      **Correct Answer:** B
14. **Which of the following statements is true about a two-dimensional array in C++?**
    - A) It can only hold integer values
    - B) It is always square-shaped
    - C) Each row can have a different number of columns
    - D) It can be used to represent matrices

      **Correct Answer:** D
15. **How do you declare a function that takes a two-dimensional array as a parameter in C++?**
    - A) `void func(int arr[maxRows][maxCols])`
    - B) `void func(int arr[])`
    - C) `void func(int **arr)`
    - D) `void func(int arr[3][3])`

      **Correct Answer:** A
16. **What does the `main` function in C++ typically return?**
    - A) `void`
    - B) `int`
    - C) `float`

- D) `char`

**Correct Answer:** B

17. **Which keyword is used to include libraries in C++?**
    - A) `import`
    - B) `include`
    - C) `require`
    - D) `using`

    **Correct Answer:** B
18. **What will the following code print if the array `a` contains values 1 to 9? `cout << a[0][0];`**
    - A) 0
    - B) 1
    - C) 9
    - D) Undefined behavior

    **Correct Answer:** B
19. **In the context of arrays, what does "iterative structure" refer to?**
    - A) A method for sorting elements
    - B) A repetitive execution of code using loops
    - C) A way to handle exceptions
    - D) A method for debugging

    **Correct Answer:** B
20. **What happens if you do not initialize an array before using it?**
    - A) It will automatically be set to zero
    - B) It may contain garbage values
    - C) It will throw a compile-time error
    - D) It will be empty

    **Correct Answer:** B
21. **Which of the following correctly represents a two-dimensional array?**
    - A) `int arr[5];`
    - B) `int arr[3][4];`
    - C) `int arr[];`
    - D) `int arr[2, 3];`

    **Correct Answer:** B
22. **How do you specify the number of columns in a two-dimensional array during declaration?**
    - A) Only in the first dimension
    - B) Only in the second dimension
    - C) In both dimensions
    - D) It is optional

    **Correct Answer:** A
23. **What does the following code do? `for (row = maxRows - 1; row >= 0; row--)`**
    - A) Iterates from the first row to the last row
    - B) Iterates from the last row to the first row
    - C) Causes an infinite loop
    - D) Initializes the rows

    **Correct Answer:** B

24. **What is the output of a `cout` statement that uses `'\n'`?**
    - A) It prints a tab
    - B) It creates a new line
    - C) It does nothing
    - D) It causes an error

    **Correct Answer:** B
25. **Which function is called to read elements into the matrix in the given code?**
    - A) `displayMatrix`
    - B) `readMatrix`
    - C) `main`
    - D) `displayFlippedMatrix`

    **Correct Answer:** B

## MCQs

1. **What is the purpose of transposing a matrix?**
    - A) To sort the elements
    - B) To interchange rows and columns
    - C) To calculate the determinant
    - D) To find the inverse
    - **Answer: B**
2. **In a square matrix of order nnn, how many elements are in the main diagonal?**
    - A) nnn
    - B) n2n^2n2
    - C) 2n2n2n
    - D) n−1n - 1n−1
    - **Answer: A**
3. **Which loop structure is typically used to iterate through a 2D array in C++?**
    - A) Single for loop
    - B) Nested for loops
    - C) While loop
    - D) Do-while loop
    - **Answer: B**
4. **If an employee's gross salary is Rs. 15,000, what is their net salary after a 10% tax deduction?**
    - A) Rs. 13,500
    - B) Rs. 14,000
    - C) Rs. 12,000
    - D) Rs. 15,000
    - **Answer: A**
5. **What is the first step in the design recipe for programming?**
    - A) Implementation
    - B) Testing
    - C) Analysis
    - D) Documentation
    - **Answer: C**

6. **In C++, which operator is used to access an array element?**
    - A) ()
    - B) []
    - C) {}
    - D) <>
    - **Answer: B**
7. **When transposing a matrix, which triangle is usually processed first?**
    - A) Lower triangle
    - B) Upper triangle
    - C) Both triangles simultaneously
    - D) None of the above
    - **Answer: B**
8. **How do you declare a two-dimensional array in C++?**
    - A) `int array[5];`
    - B) `int array[5][5];`
    - C) `int array(5, 5);`
    - D) `int array[5, 5];`
    - **Answer: B**
9. **What does the variable `temp` typically store during matrix transposition?**
    - A) The final transposed matrix
    - B) A single element from the matrix
    - C) The size of the matrix
    - D) The sum of all elements
    - **Answer: B**
10. **If a tax bracket is defined as 5% for salaries from Rs. 5,001 to Rs. 10,000, what tax applies to a salary of Rs. 7,500?**
    - A) Rs. 375
    - B) Rs. 250
    - C) Rs. 500
    - D) Rs. 600
    - **Answer: A**
11. **What will happen if you try to access an array element out of its defined bounds?**
    - A) A runtime error occurs
    - B) The program will compile but may crash
    - C) The value will be zero
    - D) The value will be undefined
    - **Answer: D**
12. **What is an identity matrix?**
    - A) A matrix with all elements as 1
    - B) A matrix where diagonal elements are 1 and others are 0
    - C) A matrix that is equal to its transpose
    - D) A matrix that can be inverted
    - **Answer: B**
13. **In the context of the provided program, which function is used to display the matrix?**
    - A) `readMatrix()`

- B) `displayMatrix()`
- C) `transposeMatrix()`
- D) `getInput()`
- **Answer: B**

14. **What type of loop is used to process each row in the matrix?**
    - A) Infinite loop
    - B) For loop
    - C) Nested loop
    - D) All of the above
    - **Answer: C**
15. **In the provided code, what data structure is used to store the gross and net salaries?**
    - A) One-dimensional array
    - B) Linked list
    - C) Two-dimensional array
    - D) Stack
    - **Answer: C**
16. **Which tax rate applies to salaries over Rs. 20,000?**
    - A) 5%
    - B) 10%
    - C) 15%
    - D) 0%
    - **Answer: C**
17. **What is the significance of the `lucky` array in the program?**
    - A) It stores gross salaries
    - B) It marks employees who have lower net salaries
    - C) It counts the total number of employees
    - D) It holds tax rates
    - **Answer: B**
18. **If a function modifies an array, what happens to the original array?**
    - A) The original array is lost
    - B) The original array remains unchanged
    - C) The original array is updated
    - D) A copy is made
    - **Answer: C**
19. **What should be the starting index for a column loop when transposing a matrix?**
    - A) 0
    - B) 1
    - C) The current row number
    - D) The number of columns
    - **Answer: C**
20. **In matrix operations, what is the result of a transpose operation on a 2D array?**
    - A) A symmetric matrix
    - B) A new matrix with rows and columns swapped
    - C) An unchanged matrix
    - D) A matrix with zero elements
    - **Answer: B**

21. **What will the `markIfUnlucky` function do in the context of employee salaries?**
    - A) Calculate total salaries
    - B) Identify employees who earn less net salary than others in lower tax brackets
    - C) Sort salaries
    - D) Print employee details
    - **Answer: B**
22. **In the context of the program, what is the output of the `printUnluckies` function?**
    - A) List of employee names
    - B) Gross salaries of all employees
    - C) Employee numbers marked as unlucky
    - D) Total salary paid by the company
    - **Answer: C**
23. **What happens to the original value of `numEmps` when passed to a function by value?**
    - A) It changes permanently
    - B) It remains the same
    - C) It becomes zero
    - D) It causes a compilation error
    - **Answer: B**
24. **Which programming paradigm does the provided solution follow?**
    - A) Functional programming
    - B) Object-oriented programming
    - C) Procedural programming
    - D) Logic programming
    - **Answer: C**
25. **What will be the output of the program if no employees are unlucky?**
    - A) A list of all employee numbers
    - B) No output related to unlucky employees
    - C) An error message
    - D) The total count of employees
    - **Answer: B**

# LECTURE 14

## MCQs on Pointers in C++

1. **What is a pointer in C++?**
    - A) A variable that stores a value
    - B) A variable that stores the address of another variable
    - C) A constant value
    - D) A data type
    - **Correct Answer: B**
2. **How do you declare a pointer to an integer in C++?**
    - A) int ptr;
    - B) int *ptr;
    - C) ptr int*;

- D) int& ptr;
- **Correct Answer: B**

3. **Which operator is used to get the address of a variable?**
   - A) *
   - B) &
   - C) #
   - D) %
   - **Correct Answer: B**
4. **What does the statement `ptr = &x;` do?**
   - A) Assigns the value of x to ptr
   - B) Assigns the address of x to ptr
   - C) Creates a pointer variable
   - D) Declares a variable x
   - **Correct Answer: B**
5. **What is the purpose of the dereferencing operator (*)?**
   - A) To declare a pointer
   - B) To get the address of a variable
   - C) To access the value at the address stored in a pointer
   - D) To assign a value to a pointer
   - **Correct Answer: C**
6. **What is a null pointer?**
   - A) A pointer that points to a valid memory address
   - B) A pointer that is not initialized
   - C) A pointer that is initialized to 0 or NULL
   - D) A pointer that points to itself
   - **Correct Answer: C**
7. **In C++, how can you pass a variable by reference to a function?**
   - A) By passing the variable itself
   - B) By passing the address of the variable
   - C) By passing a copy of the variable
   - D) By using a global variable
   - **Correct Answer: B**
8. **What is the correct way to declare a pointer to a double variable?**
   - A) double *p;
   - B) p double*;
   - C) double p*;
   - D) p = double*;
   - **Correct Answer: A**
9. **What will `*ptr` return if ptr is a pointer to an integer with the value 10?**
   - A) 10
   - B) &ptr
   - C) The address of ptr
   - D) An error
   - **Correct Answer: A**
10. **How can you initialize a pointer to a variable?**
    - A) ptr = 0;

- B) ptr = NULL;
- C) ptr = &x;
- D) All of the above
- **Correct Answer: D**

11. **What happens if you dereference a null pointer?**
    - A) It returns 0
    - B) It leads to undefined behavior
    - C) It returns a valid address
    - D) It compiles without errors
    - **Correct Answer: B**
12. **Which of the following statements is true?**
    - A) Pointers cannot be reassigned
    - B) Pointers can only point to integer types
    - C) A pointer can point to any data type
    - D) Pointers are always initialized to 0
    - **Correct Answer: C**
13. **What will happen if you pass a pointer to a function and modify the value it points to?**
    - A) The value will not change in the calling function
    - B) The value will change in the calling function
    - C) The program will crash
    - D) The pointer will become null
    - **Correct Answer: B**
14. **What is the output of the following code?**

```
cpp
Copy code
int x = 5;
int* ptr = &x;
*ptr = 10;
cout << x;
```

    - A) 5
    - B) 10
    - C) An error
    - D) Undefined behavior
    - **Correct Answer: B**
15. **Which of the following is NOT a valid pointer declaration?**
    - A) int *ptr1;
    - B) double *ptr2;
    - C) char *ptr3;
    - D) int ptr4*;
    - **Correct Answer: D**
16. **What is the primary use of pointers in C++?**
    - A) To store multiple values
    - B) To reference variables indirectly
    - C) To declare global variables
    - D) To define data types

- **Correct Answer: B**

17. **What does the expression `&(*ptr)` evaluate to?**
    - A) The value at the pointer
    - B) The address of the pointer itself
    - C) The address stored in the pointer
    - D) An error
    - **Correct Answer: C**
18. **What is the correct way to pass a pointer to a function?**
    - A) FunctionName(ptr);
    - B) FunctionName(&ptr);
    - C) FunctionName(*ptr);
    - D) FunctionName(ptr&);
    - **Correct Answer: A**
19. **Which statement is correct regarding pointer arithmetic?**
    - A) You can add any integer to a pointer
    - B) You can only subtract pointers
    - C) Pointer arithmetic is not allowed in C++
    - D) Pointers can only point to arrays
    - **Correct Answer: A**
20. **If `x` is an integer, what is the type of `int *ptr = &x;`?**
    - A) int
    - B) int*
    - C) &int
    - D) None of the above
    - **Correct Answer: B**
21. **What will the following code snippet output?**

```
cpp
Copy code
int a = 20;
int* ptr = &a;
cout << *ptr + 5;
```

    - A) 20
    - B) 25
    - C) 5
    - D) An error
    - **Correct Answer: B**
22. **In C++, what will `delete ptr;` do?**
    - A) It will remove the pointer variable
    - B) It will free the memory allocated to the pointer
    - C) It will reset the pointer to null
    - D) It will have no effect
    - **Correct Answer: B**
23. **What does the following function do?**

```
cpp
Copy code
```

```
void swap(int *a, int *b) {
    int temp = *a;
    *a = *b;
    *b = temp;
}
```

- A) Swaps the addresses of a and b
- B) Swaps the values pointed to by a and b
- C) Does nothing
- D) Causes an error
- **Correct Answer: B**

24. **Which of the following is true about pointers?**
    - A) Pointers can be initialized to any integer
    - B) Pointers must point to the same data type
    - C) Pointers can be converted to any type
    - D) Pointers can only point to global variables
    - **Correct Answer: B**
25. **What is the output of the following code?**

```
cpp
Copy code
int x = 1;
int *p = &x;
x++;
cout << *p;
```

- A) 0
- B) 1
- C) 2
- D) Undefined
- **Correct Answer: C**

## Multiple-Choice Questions (MCQs)

1. **What is a pointer in C++?**
    - A) A variable that holds a constant value
    - B) A variable that holds the address of another variable
    - C) A variable that holds a string
    - D) A data type for integers
    - **Correct Answer:** B
2. **How do you declare a pointer to an integer in C++?**
    - A) int *ptr;
    - B) int ptr*;
    - C) ptr int*;
    - D) int ptr;
    - **Correct Answer:** A
3. **What does the address operator (&) do?**
    - A) Returns the value at the address

- B) Returns the address of a variable
- C) Declares a pointer
- D) Creates a new variable
- **Correct Answer:** B

4. **What operator is used to dereference a pointer?**
   - A) &
   - B) *
   - C) %
   - D) @
   - **Correct Answer:** B
5. **In the statement `ptr = &x;`, what is being assigned to ptr?**
   - A) The value of x
   - B) The memory address of x
   - C) The size of x
   - D) The data type of x
   - **Correct Answer:** B
6. **Which of the following is NOT a correct way to initialize a pointer?**
   - A) int *ptr = NULL;
   - B) int *ptr = &x;
   - C) int *ptr = 0;
   - D) int *ptr = 10;
   - **Correct Answer:** D
7. **What does the `swap` function do in the bubble sort program?**
   - A) It initializes the array
   - B) It sorts the array in ascending order
   - C) It interchanges two values using pointers
   - D) It displays the array
   - **Correct Answer:** C
8. **What happens in a call by value?**
   - A) The actual variable is passed to the function
   - B) A copy of the variable is passed to the function
   - C) The address of the variable is passed
   - D) The variable is not passed at all
   - **Correct Answer:** B
9. **What keyword is used to define a constant pointer in C++?**
   - A) static
   - B) const
   - C) final
   - D) immutable
   - **Correct Answer:** B
10. **What is the output of the bubble sort program if the input array is {5, 3, 8, 6, 2}?**
    - A) 2, 3, 5, 6, 8
    - B) 8, 6, 5, 3, 2
    - C) 5, 3, 2, 6, 8
    - D) 6, 8, 5, 3, 2
    - **Correct Answer:** A

11. **What does the statement `*ptr = 20;` do if `ptr` points to `x`?**
    - A) It assigns the value 20 to ptr
    - B) It assigns the value 20 to x
    - C) It causes a syntax error
    - D) It returns the address of x
    - **Correct Answer:** B
12. **Which function prototype correctly indicates a function that receives a pointer as a parameter?**
    - A) void func(int);
    - B) void func(int *);
    - C) void func(*int);
    - D) void func(int &);
    - **Correct Answer:** B
13. **When passing an array to a function, what is actually passed?**
    - A) A copy of the entire array
    - B) The starting address of the array
    - C) The size of the array
    - D) The last element of the array
    - **Correct Answer:** B
14. **What is the significance of using `const` with a pointer?**
    - A) It prevents the pointer from being changed
    - B) It prevents the value at the address from being changed
    - C) It allows for dynamic memory allocation
    - D) It allows for pointer arithmetic
    - **Correct Answer:** B
15. **What is the output of the function `toupper` if the input is 'a'?**
    - A) 'a'
    - B) 'A'
    - C) 97
    - D) 65
    - **Correct Answer:** B
16. **In the context of pointers, what does 'dereferencing' mean?**
    - A) Accessing the address of a pointer
    - B) Getting the value stored at the memory address a pointer points to
    - C) Assigning a value to a pointer
    - D) Changing the address a pointer holds
    - **Correct Answer:** B
17. **Which of the following correctly initializes a pointer to the first element of an array `arr`?**
    - A) int *ptr = arr;
    - B) int *ptr = &arr[0];
    - C) int *ptr = arr[0];
    - D) Both A and B
    - **Correct Answer:** D
18. **What will happen if you dereference a null pointer?**
    - A) The program will compile successfully

- B) It will cause a runtime error
- C) It will return 0
- D) It will return the address of the pointer
- **Correct Answer:** B

19. **How do you call a function that swaps two integers using pointers?**
    - A) swap(x, y);
    - B) swap(&x, &y);
    - C) swap(x &y);
    - D) swap(x *y);
    - **Correct Answer:** B
20. **Which of the following statements about pointers is true?**
    - A) Pointers can only point to integers
    - B) A pointer can hold the address of any data type
    - C) Pointers cannot be used in arithmetic
    - D) Pointers automatically initialize to 0
    - **Correct Answer:** B
21. **What does the statement `ptr = NULL;` signify?**
    - A) ptr holds an invalid memory address
    - B) ptr points to a memory location containing 0
    - C) ptr is uninitialized
    - D) ptr points to the first element of the array
    - **Correct Answer:** A
22. **In the bubble sort implementation, what condition is checked to decide if a swap is needed?**
    - A) if (x[j] < x[j+1])
    - B) if (x[j] == x[j+1])
    - C) if (x[j] > x[j+1])
    - D) if (x[j] <= x[j+1])
    - **Correct Answer:** C
23. **What is the main benefit of using pointers in functions?**
    - A) They make the code easier to read
    - B) They allow functions to modify the original variable
    - C) They increase the speed of the program
    - D) They simplify variable declarations
    - **Correct Answer:** B
24. **What will happen if you attempt to access an array out of its bounds?**
    - A) The program will terminate gracefully
    - B) It will lead to undefined behavior
    - C) It will return zero
    - D) It will compile with warnings
    - **Correct Answer:** B
25. **In the function prototype `void convertToUppercase(char *sptr);`, what does `sptr` represent?**
    - A) A string
    - B) A character
    - C) A pointer to a character array (string)

- D) An integer
- **Correct Answer:** C

# LECTURE 15

## MCQs on Pointers and Arrays

1. **What is a pointer in C++?**
   - A) A variable that stores a value
   - B) A variable that stores the address of another variable
   - C) A type of function
   - D) None of the above

   **Answer:** B
2. **What does the name of an array represent in memory?**
   - A) The value of the first element
   - B) The address of the last element
   - C) The address of the first element
   - D) The size of the array

   **Answer:** C
3. **Which of the following is true about array names?**
   - A) They can be reassigned to another address
   - B) They are constant pointers
   - C) They can hold multiple data types
   - D) They are similar to variables

   **Answer:** B
4. **How do you access the third element of an integer array `int arr[5]`?**
   - A) arr[3]
   - B) arr(3)
   - C) *(arr + 3)
   - D) Both A and C

   **Answer:** D
5. **What is the output of `int* ptr; ptr = arr; ptr++;` if `arr` is an integer array?**
   - A) It points to the first element
   - B) It points to the second element
   - C) It causes a compilation error
   - D) It points to a random memory address

   **Answer:** B
6. **What happens when you increment a pointer?**
   - A) It increases by 1 byte
   - B) It increases by the size of the data type it points to
   - C) It resets to zero
   - D) It causes a segmentation fault

   **Answer:** B
7. **What does the expression `*ptr` mean?**
   - A) The address of ptr

- B) The value stored at the address ptr points to
- C) The size of ptr
- D) The next address in memory

**Answer:** B

8. **If `int arr[10];` is declared, what is the size of `arr`?**
   - A) 10 bytes
   - B) 20 bytes
   - C) 40 bytes (assuming int is 4 bytes)
   - D) 0 bytes

   **Answer:** C
9. **Which operator is used to get the address of a variable?**
   - A) *
   - B) &
   - C) @
   - D) #

   **Answer:** B
10. **What will be printed by `cout << &arr[0];`?**
    - A) The value of the first element
    - B) The address of the first element
    - C) The size of the array
    - D) An error

    **Answer:** B
11. **Can pointers be compared using relational operators?**
    - A) Yes
    - B) No
    - C) Only for integers
    - D) Only for arrays

    **Answer:** A
12. **What does `yptr = &arr[2];` do?**
    - A) Assigns the value of the third element to yptr
    - B) Assigns the address of the third element to yptr
    - C) Assigns the address of the second element to yptr
    - D) Causes an error

    **Answer:** B
13. **What does `yptr++` do?**
    - A) Increments the value pointed to by yptr
    - B) Increments the address held by yptr by the size of its data type
    - C) Resets yptr to zero
    - D) Causes a segmentation fault

    **Answer:** B
14. **Which of the following can be used to access the fifth element of an array `int arr[10];`?**
    - A) arr[4]
    - B) *(arr + 5)
    - C) arr + 4

- D) Both A and B

**Answer:** D

15. **What happens if you try to access an array out of bounds?**
    - A) Compilation error
    - B) It will return 0
    - C) It may return garbage value
    - D) It will cause the program to exit

    **Answer:** C
16. **Which statement is used to declare a pointer to an integer?**
    - A) int ptr;
    - B) int* ptr;
    - C) pointer int ptr;
    - D) int ptr*;

    **Answer:** B
17. **What is the output of `cout << *yptr;` if `yptr` points to an array element?**
    - A) The address of the element
    - B) The value of the element
    - C) An error
    - D) The index of the element

    **Answer:** B
18. **Which of the following statements is true regarding pointer arithmetic?**
    - A) You can add two pointers
    - B) You can subtract two pointers
    - C) You can multiply two pointers
    - D) You can divide two pointers

    **Answer:** B
19. **When a pointer is incremented, what does it actually do?**
    - A) Increments by 1 byte
    - B) Increments by the size of the data type it points to
    - C) Resets the pointer to zero
    - D) Points to the next integer only

    **Answer:** B
20. **What is the effect of dereferencing a null pointer?**
    - A) It returns zero
    - B) It causes a segmentation fault
    - C) It prints the address
    - D) It returns the value stored in the pointer

    **Answer:** B
21. **What is the primary use of pointers in programming?**
    - A) To increase variable values
    - B) To manipulate memory directly
    - C) To create arrays
    - D) To speed up arithmetic operations

    **Answer:** B
22. **Which statement correctly assigns the address of a variable `x` to a pointer `p`?**
    - A) p = x;

- B) p = &x;
- C) p = *x;
- D) &p = x;

**Answer:** B

23. **What does `*(&x)` return?**
    - A) The value of x
    - B) The address of x
    - C) An error
    - D) A pointer to x

    **Answer:** A
24. **Which of the following is NOT true about pointers?**
    - A) Pointers can point to any data type
    - B) Pointers can point to functions
    - C) Pointers cannot be compared
    - D) Pointers can be incremented

    **Answer:** C
25. **In pointer arithmetic, what is the result of `ptr1 - ptr2` if both pointers point to elements of the same array?**
    - A) The difference in their addresses
    - B) The number of elements between the two pointers
    - C) The product of their values
    - D) An error

    **Answer:** B

## MCQs on Pointers, Strings, and Arrays

1. **What is the purpose of the null character `'\0'` in C-style strings?**
    - A) To terminate the string
    - B) To indicate the start of the string
    - C) To represent a space
    - D) To count the number of characters
    - **Correct Answer: A**
2. **Which of the following is a correct way to declare a character array?**
    - A) `char name[20];`
    - B) `char name = "Amir";`
    - C) `char name[] = 'Amir';`
    - D) `char name[20] = 'Amir';`
    - **Correct Answer: A**
3. **What happens if you try to print a character array without a null terminator?**
    - A) It prints only the first character
    - B) It causes a compilation error
    - C) It continues to print until a null character is found
    - D) It prints nothing
    - **Correct Answer: C**
4. **How do you initialize a character array with a string?**
    - A) `char name[20] = "Amir";`

- B) `char name[20]; name = "Amir";`
- C) `char name[] = "Amir";`
- D) Both A and C
- **Correct Answer: D**

5. **What is the output of the following code snippet?**

```
cpp
Copy code
char name[10] = "John";
cout << name[2];
```

- A) J
- B) o
- C) h
- D) n
- **Correct Answer: C**

6. **In the context of pointers, what does the expression `*ptr` do?**
   - A) Assigns a value to ptr
   - B) Dereferences ptr to access the value it points to
   - C) Increments the pointer
   - D) None of the above
   - **Correct Answer: B**
7. **Which of the following correctly copies a string using pointers?**

```
cpp
Copy code
void myStringCopy(char *dest, const char *source) {
    // Code here
}
```

- A) `*dest = *source;`
- B) `*dest++ = *source++;`
- C) `dest = source;`
- D) `dest++ = source++;`
- **Correct Answer: B**

8. **What is the correct way to compare two integers using pointers?**
   - A) `if (xptr == yptr)`
   - B) `if (*xptr > *yptr)`
   - C) `if (x > y)`
   - D) `if (xptr > yptr)`
   - **Correct Answer: B**
9. **What is the size of an `int` on most systems?**
   - A) 1 byte
   - B) 2 bytes
   - C) 4 bytes
   - D) 8 bytes
   - **Correct Answer: C**
10. **What will happen if you forget to add the null terminator to a string?**

- A) It will work fine
- B) It can lead to undefined behavior
- C) It will throw a runtime error
- D) It will be automatically added by the compiler
- **Correct Answer: B**

11. **Which of the following statements is true regarding arrays?**
    - A) Arrays can be resized dynamically
    - B) The name of an array is a constant pointer
    - C) You cannot pass arrays to functions
    - D) Arrays do not occupy contiguous memory
    - **Correct Answer: B**
12. **What does the following code output?**

```
cpp
Copy code
char str[] = "Hello";
cout << str[4];
```

    - A) H
    - B) e
    - C) l
    - D) o
    - **Correct Answer: D**
13. **Which of the following is NOT a valid pointer declaration?**
    - A) `int *ptr;`
    - B) `char *ptr;`
    - C) `float ptr;`
    - D) `double *ptr;`
    - **Correct Answer: C**
14. **What is the purpose of the `const` keyword in function parameters?**
    - A) To make the variable mutable
    - B) To indicate that the parameter cannot be modified
    - C) To create a pointer
    - D) To increase performance
    - **Correct Answer: B**
15. **Which of the following will cause a segmentation fault?**
    - A) Accessing an array out of bounds
    - B) Dereferencing a null pointer
    - C) Both A and B
    - D) None of the above
    - **Correct Answer: C**
16. **How is memory allocated for arrays?**
    - A) Dynamically
    - B) Statically
    - C) Both A and B
    - D) Only at runtime
    - **Correct Answer: C**

17. **Which statement about pointers is FALSE?**
    - A) Pointers can point to any data type
    - B) Pointers can be incremented or decremented
    - C) Pointers can store any data type value
    - D) Pointers require initialization
    - **Correct Answer: C**
18. **What will the following code output?**

```
cpp
Copy code
int arr[5] = {1, 2, 3, 4, 5};
cout << *(arr + 2);
```

    - A) 1
    - B) 2
    - C) 3
    - D) 4
    - **Correct Answer: C**
19. **What does the expression `ptr++` do?**
    - A) Resets ptr to 0
    - B) Moves ptr to the next memory location based on its type
    - C) Dereferences ptr
    - D) None of the above
    - **Correct Answer: B**
20. **What is the correct syntax for declaring a function that copies an integer array?**
    - A) `void copyIntArray(int *dest, int *source);`
    - B) `void copyIntArray(int dest[], int source[]);`
    - C) Both A and B
    - D) None of the above
    - **Correct Answer: C**
21. **What is the output of the following code?**

```
cpp
Copy code
char str[5] = "ABC";
cout << str[3];
```

    - A) A
    - B) B
    - C) C
    - D) (Empty character)
    - **Correct Answer: D**
22. **In the context of memory, what does "contiguous" mean?**
    - A) Non-sequential
    - B) Sequential and adjacent
    - C) Random access
    - D) Dynamic allocation
    - **Correct Answer: B**

23. **Which function copies a string from source to destination using arrays?**

```
cpp
Copy code
void myStringCopy(char dest[], char source[]) {
    // Code here
}
```

   - A) `while (source[i] != '\0')`
   - B) `dest[i] = source[i];`
   - C) `dest[i] = '\0';`
   - D) Both A and B
   - **Correct Answer: D**

24. **How do you access the first element of an integer array called `arr`?**
   - A) `arr[1]`
   - B) `arr[0]`
   - C) `*arr`
   - D) Both B and C
   - **Correct Answer: D**

25. **What is the size of a `char` in C++?**
   - A) 1 byte
   - B) 2 bytes
   - C) 4 bytes
   - D) 8 bytes
   - **Correct Answer: A**

# LECTURE 16

## Multiple Choice Questions (MCQs)

1. **What is a pointer in C++?**
   - A) A variable that stores a value
   - B) A variable that stores a memory address
   - C) A constant value
   - D) A data type
   - **Correct Answer: B**
2. **Which of the following correctly declares a pointer to an integer?**
   - A) int *ptr;
   - B) int ptr*;
   - C) pointer int ptr;
   - D) int &ptr;
   - **Correct Answer: A**
3. **What does the expression `*ptr` mean when `ptr` is a pointer?**
   - A) The address stored in `ptr`
   - B) The value at the address stored in `ptr`
   - C) The size of `ptr`

- D) The type of `ptr`
- **Correct Answer: B**

4. **In C++, what is the size of a pointer?**
   - A) Depends on the data type it points to
   - B) Always 2 bytes
   - C) Always 4 bytes
   - D) Always 8 bytes
   - **Correct Answer: A**
5. **Which of the following is true about arrays and pointers?**
   - A) An array can change its address during execution.
   - B) The name of an array is a constant pointer.
   - C) Pointers cannot be used with arrays.
   - D) An array can be reassigned to a different address.
   - **Correct Answer: B**
6. **How do you access the third element of an array named `arr`?**
   - A) arr[3]
   - B) arr(2)
   - C) *arr + 2
   - D) *(arr + 3)
   - **Correct Answer: A**
7. **What is the correct way to declare a two-dimensional array of integers?**
   - A) int array[10][20];
   - B) int array(10, 20);
   - C) int array[10, 20];
   - D) int array{10, 20};
   - **Correct Answer: A**
8. **Which statement correctly increments a pointer `p` that points to an integer?**
   - A) p++;
   - B) p = p + 1;
   - C) Both A and B
   - D) p += sizeof(int);
   - **Correct Answer: C**
9. **How do you access the element at the second row and third column of a 2D array `multi`?**
   - A) multi[2][3]
   - B) multi[1][2]
   - C) *(multi + 1) + 2
   - D) *((multi + 1) + 2)
   - **Correct Answer: B**
10. **What is `argv` in the `main(int argc, char **argv)` function?**
    - A) Number of arguments
    - B) Array of character strings
    - C) A pointer to an integer
    - D) A constant value
    - **Correct Answer: B**
11. **What does `argc` represent?**

- A) The first argument
- B) Total number of arguments
- C) A pointer to the program name
- D) The last argument
- **Correct Answer: B**

12. **How do you convert a string to an integer from `argv`?**
    - A) atoi(argv[i])
    - B) int(argv[i])
    - C) convert(argv[i])
    - D) str2int(argv[i])
    - **Correct Answer: A**
13. **Which of the following correctly dereferences a pointer?**
    - A) &ptr
    - B) ptr[]
    - C) *ptr
    - D) ptr.*
    - **Correct Answer: C**
14. **In a multi-dimensional array, what does `multi + 1` do?**
    - A) Moves to the second column
    - B) Moves to the second row
    - C) Moves to the next element
    - D) It's invalid
    - **Correct Answer: B**
15. **What is the output of `cout << argc;` if no command line arguments are passed?**
    - A) 0
    - B) 1
    - C) It will cause an error
    - D) Undefined behavior
    - **Correct Answer: B**
16. **What happens if you try to access an array out of its bounds?**
    - A) It will return 0
    - B) It will cause a compile-time error
    - C) It may cause undefined behavior
    - D) It will print a warning
    - **Correct Answer: C**
17. **Which keyword is used for dynamic memory allocation in C++?**
    - A) malloc
    - B) allocate
    - C) new
    - D) create
    - **Correct Answer: C**
18. **What does the `delete` operator do?**
    - A) Allocates memory
    - B) Deallocates memory
    - C) Deletes an array
    - D) Prints a message

- **Correct Answer: B**

19. **Which of the following is a valid pointer declaration?**
    - A) int *ptr;
    - B) int ptr*;
    - C) *int ptr;
    - D) ptr int*;
    - **Correct Answer: A**
20. **How can you initialize an array of pointers to characters?**
    - A) char *arr[5];
    - B) char arr*[] = {"Hello", "World"};
    - C) char arr[5*];
    - D) char **arr = {"Hello", "World"};
    - **Correct Answer: A**
21. **What is the purpose of the `const` keyword in pointer declarations?**
    - A) To prevent pointer arithmetic
    - B) To make the pointer constant
    - C) To make the data pointed to constant
    - D) To restrict memory allocation
    - **Correct Answer: C**
22. **What does `*multi` return for a two-dimensional array `multi`?**
    - A) The first element of the array
    - B) The address of the first row
    - C) The entire array
    - D) The size of the array
    - **Correct Answer: B**
23. **How do you access the first character of a string `str` using a pointer?**
    - A) *str
    - B) str[0]
    - C) *(str + 0)
    - D) All of the above
    - **Correct Answer: D**
24. **What will `delete[] arr;` do if `arr` is a dynamically allocated array?**
    - A) Deletes a single element
    - B) Deletes the array and frees memory
    - C) Causes a runtime error
    - D) Does nothing
    - **Correct Answer: B**
25. **Which of the following is a correct way to declare an array of strings?**
    - A) char *strings[10];
    - B) char strings[10][20];
    - C) Both A and B
    - D) string strings[10];
    - **Correct Answer: C**
26. **What is the result of the expression `sizeof(arr)` for an array `int arr[10];`?**
    - A) 10
    - B) 40 (on a typical 32-bit system)

- C) 4
- D) Undefined
- **Correct Answer: B**

27. **How do you pass a multi-dimensional array to a function?**
    - A) By using the first dimension only
    - B) By using all dimensions
    - C) By passing a pointer to the first element
    - D) Both A and C
    - **Correct Answer: D**
28. **In the context of command-line arguments, what does `strcmp` function do?**
    - A) Compares two integers
    - B) Compares two strings
    - C) Converts strings to integers
    - D) Sorts an array of strings
    - **Correct Answer: B**
29. **What will happen if you try to delete a pointer that has not been dynamically allocated?**
    - A) It will cause undefined behavior
    - B) It will delete the pointer
    - C) It will do nothing
    - D) It will free up memory
    - **Correct Answer: A**
30. **What does the expression `(*ptr)++` do if `ptr` is a pointer to an integer?**
    - A) Increments the pointer
    - B) Increments the value pointed to by the pointer
    - C) Increments the address of the pointer
    - D) Decrements the value pointed to by the pointer
    - **Correct Answer: B**

## Multiple Choice Questions (MCQs)

1. **What does the `srand()` function do in C++?**
    - A) Generates a random number
    - B) Initializes the random number generator
    - C) Shuffles an array
    - D) Seeds the random number generator with a fixed value
    - **Correct Answer: B**
2. **Which function is used to generate a random number?**
    - A) random()
    - B) rand()
    - C) randNum()
    - D) generate()
    - **Correct Answer: B**
3. **What does the expression `rand() % 4` do?**
    - A) Generates a random number between 0 and 3
    - B) Generates a random number between 1 and 4

- C) Generates a random number between 0 and 4
- D) Generates a random number between -3 and 3
- **Correct Answer: A**

4. **In the context of card dealing, what does the `shuffle` function do?**
   - A) Deals cards to players
   - B) Randomizes the order of the deck
   - C) Initializes the deck
   - D) Displays the cards
   - **Correct Answer: B**
5. **What does the `deal` function do in the card program?**
   - A) Shuffles the deck
   - B) Prints the dealt cards
   - C) Creates the deck
   - D) Collects the cards
   - **Correct Answer: B**
6. **How many cards are there in a standard deck used in the program?**
   - A) 40
   - B) 52
   - C) 54
   - D) 48
   - **Correct Answer: B**
7. **Which of the following correctly initializes a deck of cards?**
   - A) int deck[52] = {0};
   - B) int deck[4][13] = {0};
   - C) int deck[4][52] = {0};
   - D) int deck[13][4] = {0};
   - **Correct Answer: B**
8. **What is the purpose of using a do-while loop in the `shuffle` function?**
   - A) To ensure the card is dealt
   - B) To ensure a unique position is chosen for each card
   - C) To loop through all cards
   - D) To shuffle the deck multiple times
   - **Correct Answer: B**
9. **Which of the following statements is true about arrays and pointers in C++?**
   - A) Array names are pointers that can change.
   - B) Pointers cannot point to arrays.
   - C) An array name acts as a constant pointer.
   - D) Pointers and arrays are completely unrelated.
   - **Correct Answer: C**
10. **What type of loop is primarily used for iterating through the rows and columns of a 2D array?**
    - A) While loop
    - B) Do-while loop
    - C) For loop
    - D) Nested loop
    - **Correct Answer: C**

11. **In the function `deal`, which parameter represents the suits of the cards?**
    - A) wDeck
    - B) wFace
    - C) wSuit
    - D) card
    - **Correct Answer: C**
12. **What is the significance of the statement `time(0)` in the context of random number generation?**
    - A) It provides a constant seed for randomization.
    - B) It ensures that the seed changes every time the program runs.
    - C) It generates a random number.
    - D) It stops the program execution.
    - **Correct Answer: B**
13. **Which of the following is the correct way to declare an array of strings for card faces?**
    - A) const char *face[52];
    - B) const char face[13][20];
    - C) const char *face[13] = {"Ace", "Two", ..., "King"};
    - D) char face[13];
    - **Correct Answer: C**
14. **How is a specific card (e.g., "Ace of Hearts") represented in the program?**
    - A) As a single string
    - B) As a combination of two strings
    - C) As an integer value only
    - D) As a character array
    - **Correct Answer: B**
15. **What will happen if you call `rand()` multiple times without calling `srand()` first?**
    - A) It will always return the same sequence of numbers.
    - B) It will generate truly random numbers.
    - C) It will crash the program.
    - D) It will produce an error message.
    - **Correct Answer: A**
16. **What is the output of `cout << card << ". " << wFace[column] << " of " << wSuit[row];` if `card` is 1, `column` is 0, and `row` is 0?**
    - A) 1. Ace of Hearts
    - B) 1. Ace of Diamonds
    - C) 1. King of Spades
    - D) 1. Ace of Clubs
    - **Correct Answer: A**
17. **In the `shuffle` function, what condition is checked before assigning a card to a position?**
    - A) If the card has already been assigned
    - B) If the position is empty
    - C) If the card is valid
    - D) If the deck is full
    - **Correct Answer: A**

18. **Which statement correctly deals with a card from the shuffled deck?**
    - A) cout << wDeck[row][column];
    - B) cout << "Dealing card: " << wDeck[row][column];
    - C) cout << wFace[card] << " of " << wSuit[row];
    - D) cout << card;
    - **Correct Answer: C**
19. **In a 2D array, what does `wDeck[row][column]` represent?**
    - A) The row number
    - B) The column number
    - C) The specific card value
    - D) The suit of the card
    - **Correct Answer: C**
20. **What happens if you run the program without including `#include <cstdlib>`?**
    - A) The program will compile successfully.
    - B) The program will not recognize `rand()` and `srand()`.
    - C) The output will be incorrect.
    - D) The program will crash.
    - **Correct Answer: B**

# LECTURE 17

## MCQs on String Handling and C Programming

1. **What header file must be included to use string manipulation functions in C?**
    - A) stdio.h
    - B) stdlib.h
    - C) string.h
    - D) ctype.h

    **Answer: C) string.h**
2. **Which function is used to copy one string to another?**
    - A) strcpy
    - B) strcopy
    - C) strncopy
    - D) copystr

    **Answer: A) strcpy**
3. **What does the function `strlen` return?**
    - A) The size of the string in bytes
    - B) The length of the string excluding the null character
    - C) The length of the string including the null character
    - D) The address of the first character in the string

    **Answer: B) The length of the string excluding the null character**
4. **Which function appends one string to another?**
    - A) strappend
    - B) strcat
    - C) strncat

- D) strmerge

**Answer: B) strcat**

5. **What is the purpose of the `atoi` function?**
   - A) Converts an integer to a string
   - B) Converts a string to an integer
   - C) Converts a float to an integer
   - D) Converts an integer to a float

   **Answer: B) Converts a string to an integer**
6. **Which function checks if a character is a digit?**
   - A) isalpha
   - B) isdigit
   - C) isspace
   - D) ispunct

   **Answer: B) isdigit**
7. **What is the ASCII value of the character 'A'?**
   - A) 65
   - B) 66
   - C) 64
   - D) 67

   **Answer: A) 65**
8. **Which function returns true if the character is an uppercase letter?**
   - A) isupper
   - B) islower
   - C) isalpha
   - D) isalnum

   **Answer: A) isupper**
9. **Which header file is necessary for character handling functions?**
   - A) stdlib.h
   - B) ctype.h
   - C) string.h
   - D) iostream.h

   **Answer: B) ctype.h**
10. **What is the result of the function `toupper('a')`?**
    - A) 'a'
    - B) 'A'
    - C) '1'
    - D) ' '

    **Answer: B) 'A'**
11. **What is the function prototype for `strncpy`?**
    - A) char *strncpy(char *s1, const char *s2, size_t n)
    - B) char *strncpy(const char *s1, const char *s2, size_t n)
    - C) void strncpy(char *s1, char *s2, int n)
    - D) char *strncpy(char *s1, char *s2, int n)

    **Answer: A) char *strncpy(char *s1, const char s2, size_t n)*
12. **What does the function `isalnum` check?**
    - A) If the character is a letter

- B) If the character is a digit
- C) If the character is alphanumeric
- D) If the character is a whitespace

**Answer: C) If the character is alphanumeric**

13. **What is returned by the `strcmp` function if two strings are equal?**
    - A) A positive number
    - B) A negative number
    - C) Zero
    - D) Null

    **Answer: C) Zero**
14. **Which of the following functions converts a string to a double?**
    - A) atof
    - B) atoi
    - C) atol
    - D) strtol

    **Answer: A) atof**
15. **Which function checks if a character is a whitespace character?**
    - A) isspace
    - B) isalpha
    - C) isdigit
    - D) ispunct

    **Answer: A) isspace**
16. **Which function returns the number of characters in a string, excluding the null character?**
    - A) strlength
    - B) strlen
    - C) countstr
    - D) getlength

    **Answer: B) strlen**
17. **What will `strcat(s1, s2)` do?**
    - A) Concatenate s1 and s2 and store in s1
    - B) Copy s2 to s1
    - C) Compare s1 and s2
    - D) Return the length of s1

    **Answer: A) Concatenate s1 and s2 and store in s1**
18. **What does `toupper(c)` return if `c` is not a lowercase letter?**
    - A) It converts `c` to uppercase
    - B) It returns `c` unchanged
    - C) It returns an error
    - D) It returns the ASCII value of `c`

    **Answer: B) It returns `c` unchanged**
19. **Which of the following is not a valid conversion function in C?**
    - A) atof
    - B) atoi
    - C) strnconv

- D) strtol

  **Answer: C) strnconv**

20. **What is the output of `isupper('b')`?**
    - A) 1
    - B) 0
    - C) 'b'
    - D) 'B'

      **Answer: B) 0**
21. **Which of the following functions can be used to check if a character is a control character?**
    - A) iscontrol
    - B) iscntrl
    - C) isalnum
    - D) isprint

      **Answer: B) iscntrl**
22. **What will `toupper('z')` return?**
    - A) 'Z'
    - B) 'A'
    - C) 'z'
    - D) '1'

      **Answer: A) 'Z'**
23. **Which function would you use to determine if a character is punctuation?**
    - A) ispunct
    - B) isprint
    - C) isgraph
    - D) isspace

      **Answer: A) ispunct**
24. **What does the function `strncat` do?**
    - A) Appends one string to another up to n characters
    - B) Copies a string up to n characters
    - C) Compares two strings up to n characters
    - D) Returns the length of a string up to n characters

      **Answer: A) Appends one string to another up to n characters**
25. **In the ASCII table, which character follows 'Z'?**
    - A) '['
    - B) 'A'
    - C) 'Y'
    - D) '0'

      **Answer: A) '['**

## MCQs on String Manipulation in C

1. **What does the function `strcpy` do?**
    - A) Concatenates two strings
    - B) Copies one string to another
    - C) Compares two strings

- D) Returns the length of a string

  **Answer: B**

2. **What is the prototype of `strncpy`?**
   - A) `char *strncpy(char *dest, const char *src, size_t n);`
   - B) `void strncpy(char *dest, char *src, int n);`
   - C) `char *strncpy(const char *dest, const char *src, size_t n);`
   - D) `char *strncpy(char *dest, const char *src);`

     **Answer: A**
3. **Which function concatenates two strings?**
   - A) `strcpy`
   - B) `strcat`
   - C) `strcmp`
   - D) `strncpy`

     **Answer: B**
4. **What does `strncat` do?**
   - A) Copies a string
   - B) Compares two strings
   - C) Concatenates a specified number of characters from one string to another
   - D) Finds the length of a string

     **Answer: C**
5. **What is the return type of `strcmp`?**
   - A) `char *`
   - B) `int`
   - C) `void`
   - D) `size_t`

     **Answer: B**
6. **Which of the following functions returns the length of a string?**
   - A) `strcpy`
   - B) `strlen`
   - C) `strcat`
   - D) `strncat`

     **Answer: B**
7. **What does `strchr` function do?**
   - A) Finds the last occurrence of a character in a string
   - B) Locates the first occurrence of a character in a string
   - C) Compares two strings
   - D) Copies a string

     **Answer: B**
8. **What does the `strstr` function do?**
   - A) Returns the length of the string
   - B) Locates the first occurrence of a substring in a string
   - C) Copies a string
   - D) Concatenates two strings

     **Answer: B**
9. **Which function is used to split a string into tokens?**
   - A) `strtok`

- B) `strcat`
- C) `strcpy`
- D) `strcmp`

**Answer: A**

10. **In `strncat(s1, s2, n)`, what does `n` represent?**
    - A) The total length of `s1`
    - B) The maximum number of characters to append from `s2`
    - C) The number of tokens in `s2`
    - D) The length of `s2`

    **Answer: B**
11. **What will `strlen("Hello")` return?**
    - A) 4
    - B) 5
    - C) 6
    - D) 0

    **Answer: B**
12. **If `s1` and `s2` are identical strings, what will `strcmp(s1, s2)` return?**
    - A) A negative number
    - B) Zero
    - C) A positive number
    - D) Undefined behavior

    **Answer: B**
13. **What is the purpose of the `const` keyword in the function prototypes?**
    - A) To allow modification of the string
    - B) To indicate the string cannot be modified
    - C) To denote a string length
    - D) To indicate a string can be passed by value

    **Answer: B**
14. **Which function would you use to find the first character not in a set of characters?**
    - A) `strcspn`
    - B) `strspn`
    - C) `strchr`
    - D) `strpbrk`

    **Answer: A**
15. **What is the effect of using `strncpy(s1, s2, 0)`?**
    - A) Copies the entire string
    - B) Copies no characters
    - C) Causes a runtime error
    - D) Appends `s2` to `s1`

    **Answer: B**
16. **Which of the following is true about `strrchr`?**
    - A) It finds the first occurrence of a character
    - B) It finds the last occurrence of a character
    - C) It concatenates strings
    - D) It compares strings

    **Answer: B**

17. **What does the `strpbrk` function return if no characters match?**
    - A) A pointer to the start of the string
    - B) NULL
    - C) The length of the string
    - D) The first character of the string

    **Answer: B**
18. **What will happen if `strcat` is used on a non-null-terminated string?**
    - A) It works correctly
    - B) Undefined behavior may occur
    - C) It copies the string correctly
    - D) It always returns an error

    **Answer: B**
19. **Which function can be used to compare the first `n` characters of two strings?**
    - A) `strcmp`
    - B) `strncmp`
    - C) `strncat`
    - D) `strcpy`

    **Answer: B**
20. **What is the maximum size of the destination array needed for `strcat`?**
    - A) Size of source string
    - B) Size of destination string
    - C) Combined size of both strings
    - D) Size of the largest string

    **Answer: C**
21. **Which function would you use to find the length of a prefix substring?**
    - A) `strcspn`
    - B) `strspn`
    - C) `strcat`
    - D) `strstr`

    **Answer: B**
22. **What happens if you pass NULL as the first argument to `strtok`?**
    - A) It resets the tokenizer
    - B) It terminates the program
    - C) It continues tokenizing the last string
    - D) It does nothing

    **Answer: C**
23. **What will `strtok` return when there are no more tokens?**
    - A) An empty string
    - B) NULL
    - C) The last token
    - D) The entire string

    **Answer: B**
24. **Which of the following is a valid statement for copying a string?**
    - A) `strcpy(s1, s2);`
    - B) `s1 = s2;`
    - C) `s1.copy(s2);`

- D) `s1.assign(s2);`

  **Answer: A**

25. **Which function can help determine if two strings are equal?**
    - A) `strncat`
    - B) `strlen`
    - C) `strcmp`
    - D) `strcpy`

      **Answer: C**

# LECTURE 18

## MCQs

1. **What is the primary purpose of file handling in programming?**
    - A) To manage memory
    - B) To perform arithmetic operations
    - C) To store data permanently
    - D) To create user interfaces
    - **Correct Answer:** C
2. **Which header file is required for file handling in C++?**
    - A) <iostream>
    - B) <fstream>
    - C) <stdlib.h>
    - D) <cstdio>
    - **Correct Answer:** B
3. **What does the `ifstream` class in C++ do?**
    - A) Writes data to files
    - B) Reads data from files
    - C) Deletes files
    - D) Creates directories
    - **Correct Answer:** B
4. **Which of the following is a correct way to open a file for reading?**
    - A) `myFile.open("filename.txt", ios::in);`
    - B) `myFile.open("filename.txt", ios::out);`
    - C) `myFile.open("filename.txt", ios::app);`
    - D) `myFile.open("filename.txt");`
    - **Correct Answer:** A
5. **What does the `ofstream` class in C++ do?**
    - A) Reads data from files
    - B) Writes data to files
    - C) Closes files
    - D) Copies files
    - **Correct Answer:** B
6. **What is the purpose of the `close()` function?**
    - A) To delete a file

- B) To open a file
- C) To close an open file
- D) To check for errors
- **Correct Answer:** C

7. **What does `myFile.eof()` check for?**
   - A) If the file exists
   - B) If the file is writable
   - C) If the end of the file has been reached
   - D) If the file is open
   - **Correct Answer:** C
8. **Which mode allows you to append data to an existing file?**
   - A) ios::in
   - B) ios::out
   - C) ios::app
   - D) ios::trunc
   - **Correct Answer:** C
9. **What will happen if you open a file with `ios::trunc` mode?**
   - A) The file will be created if it doesn't exist.
   - B) The file contents will be preserved.
   - C) The file will be emptied if it exists.
   - D) The file will be opened for reading.
   - **Correct Answer:** C
10. **What is the correct syntax to read a word from a file using an `ifstream` object?**
    - A) `myFile.get(c);`
    - B) `myFile >> c;`
    - C) `getline(myFile, c);`
    - D) `myFile.read(c);`
    - **Correct Answer:** B
11. **Which function is used to write data to a file with `ofstream`?**
    - A) `write()`
    - B) `put()`
    - C) `<<`
    - D) `writeToFile()`
    - **Correct Answer:** C
12. **In which mode can you read and write to the same file?**
    - A) ios::in
    - B) ios::out
    - C) ios::app
    - D) ios::in | ios::out
    - **Correct Answer:** D
13. **What does the `get()` function do?**
    - A) Reads an entire line
    - B) Reads a character from a file
    - C) Writes a character to a file
    - D) Reads a word from a file
    - **Correct Answer:** B

14. **What will happen if you try to read from a file that is not open?**
    - A) The program will continue without error.
    - B) The file will be automatically opened.
    - C) An error will occur.
    - D) The data will be read as empty.
    - **Correct Answer:** C
15. **Which of the following is true about `eof()` function?**
    - A) It returns true if the file is not open.
    - B) It returns true if the end of the file is reached.
    - C) It returns false if there are no errors.
    - D) It is used for writing to files.
    - **Correct Answer:** B
16. **What is the output of `cout << "File opened successfully" << endl;` if a file is opened correctly?**
    - A) The program crashes.
    - B) A success message is displayed.
    - C) No output is shown.
    - D) The file is deleted.
    - **Correct Answer:** B
17. **Which statement correctly handles a file that fails to open?**
    - A) `exit(0);`
    - B) `exit(1);`
    - C) `return;`
    - D) `continue;`
    - **Correct Answer:** B
18. **How do you check if an input file stream has encountered an error?**
    - A) `if (myFile.fail())`
    - B) `if (!myFile)`
    - C) `if (myFile.good())`
    - D) Both A and B
    - **Correct Answer:** D
19. **What does the following line of code do? `outFile.open("output.txt", ios::app);`**
    - A) Creates a new file.
    - B) Appends data to an existing file.
    - C) Opens the file for reading.
    - D) Deletes the file.
    - **Correct Answer:** B
20. **Which statement is true regarding the file system?**
    - A) Files can only be stored in the root directory.
    - B) The file system is a hierarchical structure.
    - C) Files cannot be organized into folders.
    - D) There is no concept of directories.
    - **Correct Answer:** B
21. **In the code `ifstream inFile("myfile.txt");`, what does `inFile` represent?**
    - A) A string variable
    - B) An object for reading files

- C) A function to open files
- D) A file name
- **Correct Answer:** B

22. **What is the maximum number of words that can be read at once using `myFile >> c1 >> c2;`?**
    - A) 1
    - B) 2
    - C) 3
    - D) Unlimited
    - **Correct Answer:** C
23. **Which method is used to write a single character to a file?**
    - A) `write()`
    - B) `put()`
    - C) `writeChar()`
    - D) `char()`
    - **Correct Answer:** B
24. **If a file is opened with `ios::ate`, what can you do?**
    - A) Write only at the beginning of the file.
    - B) Write only at the end of the file.
    - C) Write at any position in the file.
    - D) Only read from the file.
    - **Correct Answer:** C
25. **What will the following line of code do? `ifstream myFile("myfile.txt");`**
    - A) Create a new file.
    - B) Open a file for writing.
    - C) Open a file for reading.
    - D) Delete a file.
    - **Correct Answer:** C

## MCQs

1. **What is the typical time taken to read data from a hard disk?**
    - A) 1 microsecond
    - B) 7 milliseconds
    - C) 1 millisecond
    - D) 100 microseconds

    **Answer:** B) 7 milliseconds
2. **Why is it more efficient to read data in larger chunks?**
    - A) It uses less memory
    - B) It reduces the number of disk accesses
    - C) It simplifies coding
    - D) It requires less processing power

    **Answer:** B) It reduces the number of disk accesses
3. **What does the `getLine()` function do?**
    - A) Reads a single character from a file
    - B) Reads a line of text from a file

- C) Writes a line of text to a file
- D) Reads an entire file at once

**Answer:** B) Reads a line of text from a file

4. **What is the default stopping character for `getLine()`?**
   - A) Space
   - B) Tab
   - C) New line
   - D) Null

   **Answer:** C) New line
5. **What is the purpose of the third argument in `getLine()`?**
   - A) To specify the buffer size
   - B) To indicate where to stop reading
   - C) To specify the file name
   - D) To provide the output format

   **Answer:** B) To indicate where to stop reading
6. **What is the role of `atoi()` in the salary calculation program?**
   - A) To read integers from a file
   - B) To convert strings to integers
   - C) To sum integers
   - D) To validate input data

   **Answer:** B) To convert strings to integers
7. **In the employee salary program, what is the delimiter used for tokenizing?**
   - A) Comma
   - B) Space
   - C) Semicolon
   - D) Tab

   **Answer:** B) Space
8. **Which library is used for file handling in C++?**
   - A) <iostream>
   - B) <string>
   - C) <fstream>
   - D) <cstdlib>

   **Answer:** C) <fstream>
9. **What happens if the input file cannot be opened?**
   - A) The program continues running
   - B) The program prints an error message and exits
   - C) The program creates a new file
   - D) The program ignores the file

   **Answer:** B) The program prints an error message and exits
10. **What is the purpose of `strtok()` in the program?**
    - A) To read lines from a file
    - B) To split strings into tokens
    - C) To concatenate strings
    - D) To sort strings

    **Answer:** B) To split strings into tokens
11. **What does the `eof()` function check?**

- A) If a file is open
- B) If the end of the file is reached
- C) If there are any errors
- D) If the file is writable

**Answer:** B) If the end of the file is reached

12. **What will the output file contain after running the salary program?**
    - A) Names of employees
    - B) Total salary of employees
    - C) Average salary
    - D) List of all employees

    **Answer:** B) Total salary of employees
13. **How should files be closed in C++?**
    - A) Automatically by the system
    - B) Using the `delete` function
    - C) With the `close()` function
    - D) No need to close files

    **Answer:** C) With the `close()` function
14. **What is the maximum character limit defined for reading lines in the example programs?**
    - A) 50
    - B) 100
    - C) 256
    - D) 512

    **Answer:** B) 100
15. **What is the output format for the login name in the second program?**
    - A) First name and middle initial
    - B) Middle initial and last name
    - C) Last name only
    - D) First name only

    **Answer:** B) Middle initial and last name
16. **Which function is used to output data to a file?**
    - A) get()
    - B) put()
    - C) << operator
    - D) read()

    **Answer:** C) << operator
17. **What is the role of `ifstream`?**
    - A) To write data to files
    - B) To read data from files
    - C) To manipulate strings
    - D) To create files

    **Answer:** B) To read data from files
18. **Which of the following is true about the `main()` function?**
    - A) It must return a value
    - B) It can have no parameters
    - C) It is the starting point of execution

- D) All of the above

  **Answer:** D) All of the above

19. **In the context of the file handling program, what is "buffering"?**
    - A) Increasing file size
    - B) Storing data temporarily for efficient access
    - C) Reducing memory usage
    - D) Encrypting file content

      **Answer:** B) Storing data temporarily for efficient access
20. **What does the `exit(1)` command do?**
    - A) Exits the program successfully
    - B) Exits the program with an error code
    - C) Restarts the program
    - D) Prints an error message

      **Answer:** B) Exits the program with an error code
21. **Which data type is used for the complete line read from the file?**
    - A) int
    - B) char[]
    - C) string
    - D) float

      **Answer:** B) char[]
22. **If you want to modify the program to use command line arguments, what do you need to change?**
    - A) Add file names as parameters
    - B) Remove input/output file handling
    - C) Change the return type of main()
    - D) Use hardcoded file names

      **Answer:** A) Add file names as parameters
23. **What will happen if the array size is smaller than the line being read?**
    - A) The program will crash
    - B) Only part of the line will be read
    - C) The entire line will be read
    - D) The line will be skipped

      **Answer:** B) Only part of the line will be read
24. **How do you concatenate strings in C++?**
    - A) Using `strcat()`
    - B) Using `+` operator
    - C) Using `append()`
    - D) All of the above

      **Answer:** D) All of the above
25. **What is the role of `ofstream`?**
    - A) To read data from files
    - B) To write data to files
    - C) To manipulate strings
    - D) To close files

      **Answer:** B) To write data to files

# LECTURE 19

## MCQs

1. **What type of file access does a sequential access file use?**
   - A) Random access
   - B) Sequential access
   - C) Direct access
   - D) None of the above

   **Correct Answer: B**
2. **Which function is used to open a file in C++?**
   - A) close()
   - B) open()
   - C) read()
   - D) write()

   **Correct Answer: B**
3. **What happens when you open a file with `ofstream` in default mode?**
   - A) The file is opened for reading.
   - B) The file contents are preserved.
   - C) The existing contents are destroyed.
   - D) None of the above.

   **Correct Answer: C**
4. **Which mode should you use to open a file without deleting its contents?**
   - A) ios::in
   - B) ios::out
   - C) ios::app
   - D) ios::trunc

   **Correct Answer: C**
5. **What does the `tellg()` function do?**
   - A) Returns the next position to write in a file.
   - B) Returns the current position in a file for reading.
   - C) Moves the file pointer to the end.
   - D) Closes the file.

   **Correct Answer: B**
6. **Which function would you use to move the file pointer to a specific position in a file?**
   - A) tellg()
   - B) seekg()
   - C) open()
   - D) close()

   **Correct Answer: B**
7. **What does the `seekp()` function do?**
   - A) Moves the pointer for reading.
   - B) Moves the pointer for writing.
   - C) Returns the current read position.

- D) Closes the file.

**Correct Answer: B**

8. **In which mode can you append data to a file?**
   - A) ios::in
   - B) ios::out
   - C) ios::app
   - D) ios::trunc

   **Correct Answer: C**
9. **What will happen if you try to read past the end of a file?**
   - A) The program will terminate.
   - B) It will read garbage values.
   - C) It will throw an exception.
   - D) The pointer will loop back to the start.

   **Correct Answer: B**
10. **How do you determine the length of a file?**
    - A) Using strlen()
    - B) Using tellg() after seeking to the end
    - C) Using sizeof()
    - D) Using open()

    **Correct Answer: B**
11. **What is the purpose of the `strtok()` function?**
    - A) To open files.
    - B) To tokenize strings.
    - C) To read files.
    - D) To write files.

    **Correct Answer: B**
12. **Which of the following is a character input stream?**
    - A) ifstream
    - B) ofstream
    - C) fstream
    - D) All of the above

    **Correct Answer: A**
13. **Which function is used to read an entire line from a file?**
    - A) readLine()
    - B) getLine()
    - C) getline()
    - D) read()

    **Correct Answer: C**
14. **What does the `ios::end` parameter in `seekg()` indicate?**
    - A) Start from the beginning of the file.
    - B) Move from the current position.
    - C) Move from the end of the file.
    - D) None of the above.

    **Correct Answer: C**
15. **What will `ifstream` return if the file does not exist?**
    - A) An empty string.

- B) A pointer to null.
- C) An error state.
- D) The file size.

**Correct Answer: C**

16. **Which command is used to close a file in C++?**
    - A) terminate()
    - B) end()
    - C) close()
    - D) finish()

    **Correct Answer: C**
17. **In the context of file handling, what does "eof" stand for?**
    - A) End of file
    - B) End of function
    - C) Exit on failure
    - D) None of the above

    **Correct Answer: A**
18. **What will happen if you try to write data to a file opened in `ios::in` mode?**
    - A) Data will be written successfully.
    - B) It will overwrite existing data.
    - C) It will produce an error.
    - D) Data will be appended.

    **Correct Answer: C**
19. **Which function would you use to read data character by character?**
    - A) get()
    - B) read()
    - C) put()
    - D) None of the above

    **Correct Answer: A**
20. **What is the main disadvantage of copying a file character by character?**
    - A) It is inefficient.
    - B) It is more complex.
    - C) It is easier to implement.
    - D) None of the above.

    **Correct Answer: A**
21. **Which command is used to append data at the end of a file?**
    - A) write()
    - B) append()
    - C) put()
    - D) seekp()

    **Correct Answer: D**
22. **What is the purpose of the `ios::trunc` mode?**
    - A) To append to the file.
    - B) To preserve existing content.
    - C) To delete existing content.
    - D) To create a new file.

    **Correct Answer: C**

23. **Which of the following functions is not related to file handling?**
    - A) open()
    - B) close()
    - C) execute()
    - D) seekg()

    **Correct Answer: C**
24. **What is the result of using `seekg(-10, ios::cur)`?**
    - A) Move forward 10 bytes.
    - B) Move backward 10 bytes.
    - C) Move to the start of the file.
    - D) It produces an error.

    **Correct Answer: B**
25. **What is a common reason for needing random access to a file?**
    - A) To read sequentially.
    - B) To modify records at specific positions.
    - C) To write data only.
    - D) None of the above.

    **Correct Answer: B**

## MCQs

1. **What is the main purpose of using random access files?**
    - A) To store data permanently
    - B) To allow random read and write operations without copying the entire file
    - C) To compress data
    - D) To encrypt data
    - **Correct Answer: B**
2. **Why is it important for records in a file to have a constant length?**
    - A) To reduce file size
    - B) To avoid data corruption
    - C) To facilitate random access without disturbing file structure
    - D) To increase readability
    - **Correct Answer: C**
3. **In the given student record example, how many bytes does each record occupy?**
    - A) 50 bytes
    - B) 40 bytes
    - C) 100 bytes
    - D) 200 bytes
    - **Correct Answer: C**
4. **What is the role of a key in a file?**
    - A) To encrypt the data
    - B) To locate a specific record quickly
    - C) To compress the file
    - D) To create backups
    - **Correct Answer: B**
5. **What function is used to move the file pointer for input operations?**

- A) tellp()
- B) seekp()
- C) seekg()
- D) read()
- **Correct Answer: C**

6. **What would happen if you try to replace a character in a string with a different length without proper handling?**
   - A) The string will be unchanged
   - B) It will cause data corruption
   - C) It will write over the next characters
   - D) It will append the new string
   - **Correct Answer: C**
7. **What does the `tellp()` function do?**
   - A) Returns the current position of the input pointer
   - B) Returns the current position of the output pointer
   - C) Moves the output pointer to a specific position
   - D) Closes the output file
   - **Correct Answer: B**
8. **What is the significance of the `ios::in || ios::out` operation when opening a file?**
   - A) It opens the file in read mode only
   - B) It opens the file in write mode only
   - C) It allows both reading and writing to the file
   - D) It compresses the file
   - **Correct Answer: C**
9. **In the context of file operations, what does 'seekg()' do?**
   - A) Moves the input pointer to a specific position
   - B) Moves the output pointer to a specific position
   - C) Reads data from a file
   - D) Writes data to a file
   - **Correct Answer: A**
10. **When using `read()` and `write()` functions, what is a necessary consideration?**
    - A) The file must be opened in binary mode
    - B) The file must be opened in text mode
    - C) The file must contain only integers
    - D) The file must be less than 1MB
    - **Correct Answer: A**
11. **What is the purpose of the `seekg(-1, ios::end)` command?**
    - A) To go to the start of the file
    - B) To go to the end of the file
    - C) To move the pointer one byte before the end
    - D) To move the pointer to the middle of the file
    - **Correct Answer: C**
12. **In a file with 100-byte records, how would you access the 23rd record directly?**
    - A) By reading all previous records
    - B) By using `seekg(2200)`
    - C) By looping through each record

- D) By using `seekg(23)`
- **Correct Answer: B**

13. **Which function is NOT used for reading data from a file?**
    - A) get()
    - B) read()
    - C) put()
    - D) seekg()
    - **Correct Answer: C**
14. **What happens if you attempt to read beyond the end of a file?**
    - A) The program crashes
    - B) You get an error message
    - C) The program reads zeros
    - D) It depends on the file type
    - **Correct Answer: B**
15. **Which of the following is an advantage of using buffered I/O with `read()` and `write()`?**
    - A) Slower processing
    - B) Increased memory usage
    - C) Fewer function calls, leading to faster performance
    - D) It only works for small files
    - **Correct Answer: C**
16. **How do you write a character array to a file using the `write()` function?**
    - A) write(array)
    - B) write(array, sizeof(array))
    - C) write(array, length)
    - D) write(array, sizeof(char))
    - **Correct Answer: B**
17. **When copying a file in reverse, which function is used to set the pointer back by one byte?**
    - A) seekg(-1)
    - B) seekg(-2, ios::cur)
    - C) tellg(-1)
    - D) tellg(-2)
    - **Correct Answer: B**
18. **What does the program's final output show when writing integers using `write()`?**
    - A) The integers are displayed as readable numbers
    - B) The integers appear as binary data
    - C) The integers are completely lost
    - D) It depends on the compiler
    - **Correct Answer: B**
19. **Which of the following statements about binary files is true?**
    - A) They can be read as text
    - B) They can store any type of data
    - C) They are always smaller than text files
    - D) They cannot be created in C++
    - **Correct Answer: B**

20. **If you want to read the remaining bytes in a file after reading chunks, which code would you use?**
    - A) `fi.read(str, remainingBytes);`
    - B) `fi.read(str);`
    - C) `fi.read(str, totalBytes);`
    - D) `fi.read();`
    - **Correct Answer: A**
21. **What is the primary benefit of using `sizeof()` when writing different data types to a file?**
    - A) It ensures correct size is used regardless of architecture
    - B) It automatically formats the data
    - C) It simplifies the code
    - D) It prevents data corruption
    - **Correct Answer: A**
22. **When copying a file, what is the maximum chunk size suggested for efficient processing?**
    - A) 256 bytes
    - B) 1000 bytes
    - C) 10,000 bytes
    - D) 1,000,000 bytes
    - **Correct Answer: C**
23. **In the sample program provided, which mode does `ofstream` use when opening a file?**
    - A) ios::in
    - B) ios::out
    - C) ios::app
    - D) ios::binary
    - **Correct Answer: B**
24. **What will be the output if you read from a file opened in binary mode and print it without any conversion?**
    - A) The readable string
    - B) The raw binary data
    - C) An error message
    - D) Nothing will be printed
    - **Correct Answer: B**
25. **Which command correctly reads data from a file until the end is reached?**
    - A) while (inputFile.get(character))
    - B) while (inputFile.read(character))
    - C) while (inputFile.read())
    - D) while (inputFile.get())
    - **Correct Answer: A**

# LECTURE 20

## MCQs on Structures and Unions

1. **What keyword is used to declare a structure in C/C++?**
    - A) class
    - B) struct
    - C) union
    - D) define

    **Answer: B) struct**
2. **Which of the following is a valid structure declaration?**
    - A) struct student { char name[60]; float GPA; };
    - B) struct { int age; } person;
    - C) struct student student1;
    - D) All of the above

    **Answer: D) All of the above**
3. **How do you access a data member of a structure using a variable?**
    - A) student.name
    - B) student->name
    - C) student.name()
    - D) student->name()

    **Answer: A) student.name**
4. **What is the purpose of the sizeof operator?**
    - A) To determine the number of members in a structure
    - B) To determine the size of a data type in bytes
    - C) To allocate memory for a structure
    - D) None of the above

    **Answer: B) To determine the size of a data type in bytes**
5. **How do you initialize a structure at the time of declaration?**
    - A) struct student s1 = {“John”, 20};
    - B) struct student s1(“John”, 20);
    - C) struct student s1[“John”, 20];
    - D) struct student s1 : {“John”, 20};

    **Answer: A) struct student s1 = {“John”, 20};**
6. **Which of the following is true about structures in C/C++?**
    - A) Structures can only contain variables of the same type.
    - B) Structures cannot contain other structures.
    - C) Structures can contain different data types.
    - D) Structures are passed by reference only.

    **Answer: C) Structures can contain different data types.**
7. **What is the correct way to declare a pointer to a structure?**
    - A) struct student *sptr;
    - B) student *sptr;
    - C) pointer struct student sptr;
    - D) Both A and B

    **Answer: D) Both A and B**
8. **How do you access a member of a structure using a pointer?**
    - A) (*ptr).member
    - B) ptr->member
    - C) Both A and B

- D) ptr.member

**Answer: C) Both A and B**

9. **Which statement correctly copies one structure to another?**
   - A) s1.copy(s2);
   - B) s1 = s2;
   - C) s1.assign(s2);
   - D) s1 << s2;

   **Answer: B) s1 = s2;**
10. **What happens if you try to add two structures using the + operator?**
    - A) It compiles without error.
    - B) It results in a syntax error.
    - C) It performs member-wise addition.
    - D) It is undefined behavior.

    **Answer: B) It results in a syntax error.**
11. **Which of the following is NOT a valid member of a structure?**
    - A) int
    - B) float
    - C) struct
    - D) pointer

    **Answer: C) struct**
12. **How do you declare an array of structures?**
    - A) struct student arr[10];
    - B) student arr[10];
    - C) Both A and B
    - D) array struct student[10];

    **Answer: C) Both A and B**
13. **What is the main difference between structures and unions?**
    - A) Structures use more memory than unions.
    - B) Unions can only hold one member at a time.
    - C) Structures can only contain integer types.
    - D) Both A and B

    **Answer: D) Both A and B**
14. **How can you pass a structure to a function?**
    - A) By reference only
    - B) By value only
    - C) Both by value and by reference
    - D) Not allowed

    **Answer: C) Both by value and by reference**
15. **Which operator is used to access members of a structure using a pointer?**
    - A) .
    - B) ->
    - C) *
    - D) &

    **Answer: B) ->**
16. **What will be the output of `sizeof(s1)` where `s1` is a structure containing two `int` and one `char[10]`?**

- A) 10 bytes
- B) 14 bytes
- C) 8 bytes
- D) Depends on the compiler

**Answer: D) Depends on the compiler**

17. **If you have a structure with an array as a member, how is that array passed to functions?**
    - A) By value
    - B) By reference
    - C) It cannot be passed
    - D) It is always passed by pointer

    **Answer: A) By value**
18. **How do you define a structure containing another structure?**
    - A) struct outer { struct inner; };
    - B) struct outer { inner in; };
    - C) struct outer { struct inner { ... }; };
    - D) Both B and C

    **Answer: D) Both B and C**
19. **Which function can be used to copy strings to a character array within a structure?**
    - A) strcpy()
    - B) copy()
    - C) assign()
    - D) append()

    **Answer: A) strcpy()**
20. **What is the purpose of using `typedef` with structures?**
    - A) To create a new data type
    - B) To define a variable
    - C) To declare functions
    - D) None of the above

    **Answer: A) To create a new data type**
21. **Can structures have members that are pointers to other structures?**
    - A) Yes
    - B) No
    - C) Only in C++
    - D) Only in C

    **Answer: A) Yes**
22. **How do you display the age of the first student in an array of structures?**
    - A) cout << s[0].age;
    - B) cout << s[1].age;
    - C) cout << s.age[0];
    - D) cout << s.age[1];

    **Answer: A) cout << s[0].age;**
23. **If you have a pointer to a structure, which of the following is correct?**
    - A) sptr.name
    - B) (*sptr).name
    - C) sptr->name

- D) Both B and C

  **Answer: D) Both B and C**

24. **Which of the following is true about the `struct` keyword?**
    - A) It is optional in C++.
    - B) It cannot be used in C.
    - C) It can only be used to define classes.
    - D) None of the above

      **Answer: A) It is optional in C++.**
25. **What happens if a structure is declared but not initialized?**
    - A) The structure contains garbage values.
    - B) The structure defaults to zero values.
    - C) It results in a compilation error.
    - D) The values are automatically set to null.

      **Answer: A) The structure contains garbage values.**

## MCQs on Structures and Unions

1. **What is the primary purpose of a structure in C/C++?**
    - A) To perform arithmetic operations
    - B) To group different data types
    - C) To handle memory management
    - D) To implement loops
    - **Correct Answer: B**
2. **Which of the following is a valid way to declare a structure?**
    - A) `struct Student { char name[30]; };`
    - B) `struct Student name[30];`
    - C) `Student { char name[30]; };`
    - D) `Student name;`
    - **Correct Answer: A**
3. **How do you access a member of a structure?**
    - A) Using the `::` operator
    - B) Using the `->` operator
    - C) Using the `.` operator
    - D) Using the `[]` operator
    - **Correct Answer: C**
4. **What will be the size of a structure containing an `int` and a `char`?**
    - A) 1 byte
    - B) 2 bytes
    - C) 4 bytes
    - D) 5 bytes
    - **Correct Answer: C**
5. **What is the syntax to define a union?**
    - A) `union MyUnion { int a; float b; };`
    - B) `MyUnion union { int a; float b; };`
    - C) `union MyUnion ( int a, float b );`
    - D) `MyUnion { int a; float b; };`

- **Correct Answer: A**

6. **Which statement is true about unions?**
   - A) All members of a union have separate memory.
   - B) A union can hold multiple values simultaneously.
   - C) The size of a union is equal to the size of its largest member.
   - D) Unions cannot contain structures.
   - **Correct Answer: C**
7. **If a structure contains a `double` and an `int`, what will be the size of the structure?**
   - A) Size of double only
   - B) Size of int only
   - C) Sum of both sizes
   - D) Maximum size of either
   - **Correct Answer: C**
8. **In a structure, how can you initialize a member?**
   - A) `struct student s = { "Ali", "CS201", 20, 3.5 };`
   - B) `student s = ( "Ali", "CS201", 20, 3.5 );`
   - C) `struct student s = new student();`
   - D) `s.name = "Ali";`
   - **Correct Answer: A**
9. **How do you declare an array of structures?**
   - A) `struct Student arr[10];`
   - B) `Student arr[10];`
   - C) `array struct Student[10];`
   - D) `struct array Student[10];`
   - **Correct Answer: A**
10. **What happens if you try to access a union member that was not the last assigned member?**
    - A) The program crashes
    - B) It may show a garbage value
    - C) It shows the last assigned value
    - D) It shows zero
    - **Correct Answer: B**
11. **Which operator is used to access a member of a structure pointer?**
    - A) `.`
    - B) `->`
    - C) `::`
    - D) `&`
    - **Correct Answer: B**
12. **If `s` is a structure variable of type `Student`, how do you assign a value to the `age` member?**
    - A) `s.age = 20;`
    - B) `s->age = 20;`
    - C) `s(age) = 20;`
    - D) `age(s) = 20;`
    - **Correct Answer: A**
13. **What is the correct way to read data into a structure from a file?**

- A) `fscanf(file, "%s", s.name);`
- B) `cin >> s.name;`
- C) `getline(file, s.name);`
- D) `file >> s.name;`
- **Correct Answer: D**

14. **Which of the following is not a valid way to return a structure from a function?**
    - A) `return struct_variable;`
    - B) `return &struct_variable;`
    - C) `return struct_type;`
    - D) `return struct_variable;`
    - **Correct Answer: B**
15. **In the context of a union, which statement is true?**
    - A) You can access all members simultaneously.
    - B) The size of the union is determined by the smallest member.
    - C) The last assigned member's value will be retained.
    - D) You can only assign values to one member at a time.
    - **Correct Answer: C**
16. **Which of the following is the correct declaration of a union that contains an integer and a float?**
    - A) `union Data { int i; float f; };`
    - B) `union Data(int i, float f);`
    - C) `Data union { int i; float f; };`
    - D) `union Data { int, float; };`
    - **Correct Answer: A**
17. **Which of the following will not compile?**
    - A) `struct Student { int age; };`
    - B) `union Data { int a; int b; };`
    - C) `int Data = 10;`
    - D) `struct Student s;`
    - **Correct Answer: C**
18. **What will the size of the union `union { char c; int i; }` be on a system where `int` is 4 bytes?**
    - A) 1 byte
    - B) 2 bytes
    - C) 4 bytes
    - D) 5 bytes
    - **Correct Answer: C**
19. **If you want to store multiple data types in the same memory location, which construct should you use?**
    - A) Structure
    - B) Array
    - C) Union
    - D) Class
    - **Correct Answer: C**
20. **When defining a structure, what keyword is used?**
    - A) `struct`

- B) `class`
- C) `union`
- D) `data`
- **Correct Answer: A**

21. **What will the output be if you print a union member after assigning a different member?**
    - A) Correct value of the last assigned member
    - B) Undefined behavior
    - C) Last assigned member's value will be printed
    - D) A compile-time error
    - **Correct Answer: A**
22. **Which function can you use to write a structure to a file?**
    - A) `fprintf()`
    - B) `fwrite()`
    - C) `fprintf()`
    - D) Both A and B
    - **Correct Answer: D**
23. **What keyword is used to define a union in C/C++?**
    - A) `class`
    - B) `struct`
    - C) `union`
    - D) `data`
    - **Correct Answer: C**
24. **Which of the following correctly initializes a union member?**
    - A) `u.i = 5;`
    - B) `u.c = 'A';`
    - C) `u.i = 5; u.c = 'A';`
    - D) All of the above
    - **Correct Answer: D**
25. **How do you find the size of a structure or union in C/C++?**
    - A) `sizeof()`
    - B) `length()`
    - C) `size()`
    - D) `getSize()`
    - **Correct Answer: A**

# LECTURE 21

## MCQs on Bit Manipulation

1. **What is the basic unit of memory?**
    - A) Byte
    - B) Bit
    - C) Kilobyte

- D) Nibble

  **Correct Answer:** B) Bit

2. **How many bits are there in a byte?**
   - A) 4
   - B) 8
   - C) 16
   - D) 32

     **Correct Answer:** B) 8
3. **Which operator is used for bitwise AND?**
   - A) &&
   - B) ||
   - C) &
   - D) |

     **Correct Answer:** C) &
4. **What does the bitwise OR operator (|) return if both bits are 0?**
   - A) 1
   - B) 0
   - C) Undefined
   - D) -1

     **Correct Answer:** B) 0
5. **Which operator returns 1 only if one input bit is 1 and the other is 0?**
   - A) AND
   - B) OR
   - C) XOR
   - D) NOT

     **Correct Answer:** C) XOR
6. **What is the result of `~1` using the NOT operator?**
   - A) 0
   - B) 1
   - C) -2
   - D) Undefined

     **Correct Answer:** C) -2
7. **Which operator can be used to set a specific bit to 1?**
   - A) &
   - B) |
   - C) ^
   - D) ~

     **Correct Answer:** B) |
8. **In the expression `number & 0x8`, what does `0x8` represent?**
   - A) Decimal 8
   - B) Binary 1000
   - C) Hexadecimal 8
   - D) All of the above

     **Correct Answer:** D) All of the above
9. **How do you determine if the fourth bit of a number is set?**
   - A) Check if `number & 0x4`

- B) Check if `number & 0x8`
- C) Check if `number & 0x2`
- D) Check if `number & 0x1`

**Correct Answer:** B) Check if `number & 0x8`

10. **What is the result of `12 | 8`?**
    - A) 12
    - B) 8
    - C) 20
    - D) 4

    **Correct Answer:** A) 12
11. **What does the expression `a ^ b ^ b` equal?**
    - A) a
    - B) b
    - C) 0
    - D) Undefined

    **Correct Answer:** A) a
12. **Which operator is a unary operator?**
    - A) &
    - B) |
    - C) ~
    - D) ^

    **Correct Answer:** C) ~
13. **What is the purpose of bit masking?**
    - A) To check if a specific bit is set
    - B) To change the value of a bit
    - C) To combine two numbers
    - D) All of the above

    **Correct Answer:** A) To check if a specific bit is set
14. **In bit manipulation, what does the left shift operator (<<) do?**
    - A) Divides by 2
    - B) Multiplies by 2
    - C) Inverts bits
    - D) None of the above

    **Correct Answer:** B) Multiplies by 2
15. **What does the right shift operator (>>) do?**
    - A) Multiplies by 2
    - B) Divides by 2
    - C) Flips bits
    - D) None of the above

    **Correct Answer:** B) Divides by 2
16. **What is the output of `cout << (8 >> 1)`?**
    - A) 8
    - B) 4
    - C) 2
    - D) 0

    **Correct Answer:** B) 4

17. **What is the result of `15 ^ 7`?**
    - A) 8
    - B) 6
    - C) 9
    - D) 0

    **Correct Answer:** C) 9
18. **What happens when you perform an AND operation on any number with 0?**
    - A) The result is 1
    - B) The result is the original number
    - C) The result is 0
    - D) Undefined

    **Correct Answer:** C) The result is 0
19. **In C++, how do you declare an unsigned integer?**
    - A) int x;
    - B) unsigned int x;
    - C) uint x;
    - D) All of the above

    **Correct Answer:** B) unsigned int x;
20. **What is a common use case for XOR in password handling?**
    - A) Compressing passwords
    - B) Encrypting and decrypting passwords
    - C) Storing passwords
    - D) None of the above

    **Correct Answer:** B) Encrypting and decrypting passwords
21. **How do you swap two numbers using XOR without a temporary variable?**
    - A) a = a + b; b = a - b; a = a - b;
    - B) a = a ^ b; b = b ^ a; a = a ^ b;
    - C) a = b; b = a;
    - D) None of the above

    **Correct Answer:** B) a = a ^ b; b = b ^ a; a = a ^ b;
22. **What is the effect of `~0`?**
    - A) 0
    - B) 1
    - C) -1
    - D) Undefined

    **Correct Answer:** C) -1
23. **Which bitwise operation can be used to clear a specific bit?**
    - A) AND with a mask of 0
    - B) OR with a mask of 1
    - C) XOR with a mask of 1
    - D) AND with a mask of 1

    **Correct Answer:** A) AND with a mask of 0
24. **How is the exclusive OR operator represented?**
    - A) &
    - B) |
    - C) ^

- D) ~

**Correct Answer:** C) ^

25. **What is the result of `~(12)` if `12` is represented in binary?**
    - A) 12
    - B) -13
    - C) 13
    - D) -12

    **Correct Answer:** B) -13
26. **What is the output of `number << 1` if `number = 5`?**
    - A) 10
    - B) 11
    - C) 5
    - D) 0

    **Correct Answer:** A) 10
27. **What does `cout << (8 & 5)` output?**
    - A) 8
    - B) 0
    - C) 4
    - D) 5

    **Correct Answer:** C) 0
28. **In a system using RAID, what is XOR used for?**
    - A) To calculate data
    - B) To recover data from lost disks
    - C) To encrypt data
    - D) None of the above

    **Correct Answer:** B) To recover data from lost disks
29. **What happens when you right shift a signed negative number?**
    - A) Always becomes positive
    - B) Retains its sign
    - C) Always becomes zero
    - D) Undefined behavior

    **Correct Answer:** B) Retains its sign
30. **Which bit manipulation operator is useful for checking if a bit is set?**
    - A) OR
    - B) AND
    - C) NOT
    - D) XOR

    **Correct Answer:** B) AND
31. **If you have a binary number `1101`, what is the result of `1101 ^ 1010`?**
    - A) 0111
    - B) 1111
    - C) 0101
    - D) 0000

    **Correct Answer:** A) 0111
32. **Which of the following is true about unsigned integers?**
    - A) Can be negative

- B) Can only be positive
- C) Can hold larger values than signed integers
- D) Both B and C
  **Correct Answer:** D) Both B and C

33. **What is the result of `~(a & b)` using De Morgan's laws?**
    - A) ~a | ~b
    - B) ~a & ~b
    - C) a | b
    - D) None of the above
      **Correct Answer:** A) ~a | ~b
34. **In C++, what is the size of an `unsigned char`?**
    - A) 1 byte
    - B) 2 bytes
    - C) 4 bytes
    - D) 8 bytes
      **Correct Answer:** A) 1 byte
35. **What happens when you perform a bitwise OR operation between two identical bits?**
    - A) The result is 1
    - B) The result is 0
    - C) The result is undefined
    - D) The result depends on the context
      **Correct Answer:** A) The result is 1

# LECTURE 22

- **What is the purpose of the += operator in C++?**
  - A) To assign a new value
  - B) To increment a variable
  - C) To compare two variables
  - D) To perform a bitwise operation
    **Correct Answer:** B) To increment a variable

- **Which operator is unary?**
  - A) &
  - B) |
  - C) ^
  - D) ~
    **Correct Answer:** D) ~

- **What data type is used to store whole numbers in C++?**
  - A) float

- B) double
- C) int
- D) char
  **Correct Answer:** C) int

- **How are arrays indexed in C++?**

  - A) 1-based indexing
  - B) 0-based indexing
  - C) Random indexing
  - D) Negative indexing
    **Correct Answer:** B) 0-based indexing

- **What will the expression `7 % 2` yield?**

  - A) 3
  - B) 4
  - C) 1
  - D) 0
    **Correct Answer:** C) 1

- **Which operator is used for bitwise AND in C++?**

  - A) &&
  - B) ||
  - C) &
  - D) |
    **Correct Answer:** C) &

- **What is the purpose of the `switch` statement?**

  - A) To loop through a set of values
  - B) To make multi-way decisions
  - C) To create a conditional statement
  - D) To exit a loop
    **Correct Answer:** B) To make multi-way decisions

- **What does the `continue` statement do in a loop?**

  - A) Terminates the loop
  - B) Exits the current function
  - C) Skips the remaining statements in the loop for the current iteration
  - D) Restarts the program
    **Correct Answer:** C) Skips the remaining statements in the loop for the current iteration

- **In which case would you use the `do-while` loop?**

- A) When you want to ensure the loop executes at least once
- B) When you need to loop a specific number of times
- C) When you want to loop indefinitely
- D) When the number of iterations is unknown
  **Correct Answer:** A) When you want to ensure the loop executes at least once

- **What is the correct syntax for an `if` statement?**

  - A) if condition { // code }
  - B) if (condition) { // code }
  - C) if { condition } // code
  - D) if (condition) // code
    **Correct Answer:** B) if (condition) { // code }

- **What does the `break` statement do?**

  - A) Exits a loop or switch statement
  - B) Continues the next iteration of a loop
  - C) Stops the program execution
  - D) Breaks a variable into parts
    **Correct Answer:** A) Exits a loop or switch statement

- **How do you declare a pointer in C++?**

  - A) int* ptr;
  - B) int ptr*;
  - C) pointer int ptr;
  - D) int& ptr;
    **Correct Answer:** A) int* ptr;

- **Which of the following is a logical operator?**

  - A) +
  - B) *
  - C) &&
  - D) |
    **Correct Answer:** C) &&

- **What will happen if you write `if (a = b)`?**

  - A) It will compare a and b
  - B) It will always evaluate to true
  - C) It will throw a compilation error
  - D) It will assign b to a
    **Correct Answer:** D) It will assign b to a

- **What does the `sizeof(int)` typically return?**

  - A) 1
  - B) 2
  - C) 4
  - D) 8
    **Correct Answer:** C) 4

- **How can you access the third element of an array named `arr`?**

  - A) arr[2]
  - B) arr[3]
  - C) arr(3)
  - D) arr{3}
    **Correct Answer:** A) arr[2]

- **Which statement is true about functions in C++?**

  - A) Functions cannot return values
  - B) Functions can call themselves
  - C) Functions must have at least one parameter
  - D) Functions cannot have parameters
    **Correct Answer:** B) Functions can call themselves

- **What will `a &= b;` do?**

  - A) Set a to the sum of a and b
  - B) Set a to the result of bitwise AND between a and b
  - C) Increment a by b
  - D) Assign b to a
    **Correct Answer:** B) Set a to the result of bitwise AND between a and b

- **What is the output of `if (x > 0) { cout << "Positive"; }` when x is -1?**

  - A) Positive
  - B) Negative
  - C) No output
  - D) Compilation error
    **Correct Answer:** C) No output

- **Which of the following is a correct way to declare an array of integers?**

  - A) int arr[10];
  - B) int arr(10);
  - C) int arr{10};

- D) int arr[10]; int arr[5];
  **Correct Answer:** A) int arr[10];

- **What does the `float` data type represent?**

  - A) Integer values
  - B) Character values
  - C) Decimal values
  - D) Boolean values
    **Correct Answer:** C) Decimal values

- **Which of the following is NOT a compound assignment operator?**

  - A) +=
  - B) -=
  - C) /=
  - D) <>
    **Correct Answer:** D) <>

- **In a nested `if` statement, how do you handle multiple conditions?**

  - A) Using logical operators like && and ||
  - B) Using only one if statement
  - C) Ignoring the second condition
  - D) Using a switch statement
    **Correct Answer:** A) Using logical operators like && and ||

- **How do you declare a character variable in C++?**

  - A) char c;
  - B) character c;
  - C) char(c);
  - D) c char;
    **Correct Answer:** A) char c;

- **What happens if you do not use a `break` statement in a switch case?**

  - A) It will skip to the next case
  - B) It will cause a compilation error
  - C) It will execute the default case
  - D) It will terminate the program
    **Correct Answer:** A) It will skip to the next case

- **Which loop executes at least once?**

  - A) for loop

- B) while loop
- C) do-while loop
- D) none of the above
  **Correct Answer:** C) do-while loop

- **What is a pointer?**

  - A) A variable that stores a value
  - B) A variable that stores the address of another variable
  - C) A type of array
  - D) A method of looping
    **Correct Answer:** B) A variable that stores the address of another variable

- **Which of the following is a correct way to open a file in C++?**

  - A) fopen("file.txt");
  - B) open("file.txt");
  - C) ifstream file("file.txt");
  - D) read("file.txt");
    **Correct Answer:** C) ifstream file("file.txt");

- **What is the main advantage of using functions in programming?**

  - A) They reduce code duplication
  - B) They increase compilation time
  - C) They make the code less readable
  - D) They are not necessary
    **Correct Answer:** A) They reduce code duplication

- **Which operator is used to compare two values for equality?**

  - A) =
  - B) ==
  - C) !=
  - D) <
    **Correct Answer:** B) ==

- **What is the initial value of an uninitialized variable in C++?**

  - A) 0
  - B) Undefined
  - C) 1
  - D) -1
    **Correct Answer:** B) Undefined

- **What will happen if you try to access an out-of-bounds index in an array?**

- A) It will return 0
- B) It will throw an error
- C) It will return garbage value
- D) It will cause a stack overflow
  **Correct Answer:** C) It will return garbage value

- **Which of the following is a correct way to comment in C++?**

  - A) // This is a comment
  - B) /* This is a comment */
  - C) -- This is a comment
  - D) Both A and B
    **Correct Answer:** D) Both A and B

- **Which keyword is used to define a constant in C++?**

  - A) const
  - B) static
  - C) final
  - D) constant
    **Correct Answer:** A) const

- **What does the `break` statement do in a switch-case?**

  - A) It terminates the program
  - B) It jumps to the end of the switch block
  - C) It skips the next case
  - D) It continues to the next case
    **Correct Answer:** B) It jumps to the end of the switch block

- **What is the range of the `char` data type?**

  - A) -128 to 127
  - B) 0 to 255
  - C) -256 to 255
  - D) 0 to 127
    **Correct Answer:** A) -128 to 127

- **What will `for (int i = 0; i < 10; i++)` do?**

  - A) Loop from 1 to 10
  - B) Loop from 0 to 9
  - C) Loop indefinitely
  - D) Cause a compilation error
    **Correct Answer:** B) Loop from 0 to 9

- **What is the default value of a boolean variable in C++?**

  - A) true
  - B) false
  - C) 1
  - D) 0
    **Correct Answer:** B) false

- **Which of the following is a C++ Standard Library for input and output?**

  - A) stdio.h
  - B) iostream
  - C) io.h
  - D) stdlib.h
    **Correct Answer:** B) iostream

- **What does `return 0;` indicate in the main function?**

  - A) An error occurred
  - B) The program terminated successfully
  - C) The program is still running
  - D) No value is returned
    **Correct Answer:** B) The program terminated successfully

**EFFORT BY VU BSCS (M.TAQI)**